

Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems

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Abstract

Current AI-mediated coordination work accumulates reasoning, decisions, and rationale across sessions and participants, but most of that coordination state is lost at session boundaries, fragmented across documents, tickets, and chat logs, or embedded implicitly in vendor-specific product schemas. This paper defines the **Coordination Knowledge Substrate (CKS)** — a design pattern in which coordination knowledge, governance semantics, and conflict-handling logic live in a persistent, human-governed substrate outside the LLM — and defends it through six architectural commitments: hybrid substrate-LLM division of labor, conflict preservation as first-class substrate state, human-governed authority, AI as substrate mediator, tool-agnosticism at the substrate layer, and linear-cost scaling inherited from database-backed external memory.

The core theory is defended through five claims drawing on systematic review of 2024–2026 literature across governance-first AI architectures, epistemically-structured substrates, and execution-substrate research. A sixth claim extends the core theory to human-AI-human coordination: the same substrate that supports a single user coordinating across sessions supports multiple roles coordinating across sessions, with AI mediating over the substrate rather than in per-session interactions. The paper demonstrates both the core theory and the extension through one proof-of-concept in a regulated industry, instantiated in commodity spreadsheet infrastructure with a commercially available LLM integration.

The contribution is a design pattern, not an empirical evaluation. CKS articulates a recurring structure of coordination problem and architectural response — named so that subsequent work can adopt it, vary it, compose it with adjacent patterns, or argue against it. The paper positions CKS as one contribution to an emerging research direction in CSCW-AI, governance-first AI architectures, and epistemically-structured substrates.

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1. Introduction

1.1 Problem framing — the coordination gap in current AI-mediated work

In a regulated industry, a multi-role coordination workflow runs across commodity tools with an LLM assistant in the loop. A participant in one role makes a decision; the LLM helps draft the justification; a participant in a second role reviews and revises it; a third, days later, needs to extend the work and reconstruct why the earlier decision went the way it did. The decision itself can be recovered — the outcome sits in a document, a ticket comment, or a chat log. The reasoning behind it is harder to recover, and the state needed to carry it forward into the next session does not exist anywhere as an addressable artifact. This is the coordination gap in current AI-mediated work: reasoning, decisions, and rationale produced across sessions and participants, then lost at session boundaries, fragmented across surfaces, and not present as coordination state that later work can address.

The gap matters because regulated and quasi-regulated coordination depends on what was decided, by whom, under what authority, and with what rationale being recoverable after the fact — not as recollection, but as structured state that subsequent participants and subsequent sessions can

inspect and act on. Current practice handles this through four partial mechanisms: documents carry narrative explanations of decisions; tickets carry outcomes; chat logs carry conversational history; the LLM’s context window carries the working state of the session during which the decision was made. Each stores part of what happened. None is a coordination artifact in the sense the paper will define — a structured, persistent, human-governed representation of decisions, rationale, authority, and unresolved conflicts, addressable across sessions and participants. The landscape recognizes the gap. The Atlan 2026 enterprise-KB framing (industry source) argues for governed institutional knowledge that survives across workflow instances; the neural-closed-box critique of vector-RAG opacity (Saxena, Gaur, Sheth 2026) names the symmetric problem at the retrieval layer — retrieval indices are opaque to the humans who need to govern what the LLM reasons over. Both describe facets of the same underlying gap.

Two adjacent framings sit close enough to the gap that a reader is likely to file any response to it under one or the other. The first is RAG-with-governance: retrieval augmented by policy, access control, and provenance metadata to make the retrieved content trustworthy. This is a retrieval approach; it improves what the LLM reads, but does not turn what was decided, by whom, and why into an addressable artifact the next participant can inspect and extend. The second is the AI-gateway and control-plane family: middleware that wraps the LLM with policy enforcement, audit logging, and request routing. Control-plane architectures govern around the runtime — which requests the LLM sees, what it is permitted to produce, what is logged — but leave the coordination knowledge behind the decisions in whatever form the application happened to produce. Neither framing treats the coordination substrate as a design object in its own right. That is what current practice is missing.

1.2 Contribution statement — CKS as a design pattern for coordination outside the model

This paper defends a design pattern for AI systems that we call the Coordination Knowledge Substrate (CKS). The contribution is architectural: in CKS, the coordination knowledge, governance semantics, and conflict-handling logic that structured work depends on live in a persistent, human-governed substrate outside the LLM — not derived inside the LLM’s inference, and not outsourced to specialized runtimes around it. The paper uses a single phrase for this move, and the phrase is used throughout: **outside-the-model vs. inside-the-model**.

The phrase invites a narrower reading than the paper intends, so one clarification lands up front. Throughout the paper, “outside the model” refers to the architectural location of coordination, governance, and decision state — in a persistent, human-governed substrate — rather than to the narrower parametric-vs-non-parametric distinction standard in retrieval-augmented generation and external-memory work. CKS is not RAG: the substrate is not an index over documents, but a coordination artifact humans read from, write into, and hold authority over.

The paper is structured around the core theory and one extension. Part 1 states and defends the core theory, which describes a single human coordinating structured work across sessions with LLM assistance over a persistent substrate; this is the minimum viable instantiation of the pattern, and it stands on its own. Part 2 presents the human-AI-human coordination extension, in which the same pattern is applied to multi-human, multi-role settings, and coordination across participants is mediated by the substrate they share. Part 3 presents the proof-of-concept (POC) as an integrated demonstration of both: a single regulated-domain implementation in commodity tools that exercises the core theory and the extension together.

Five disambiguations follow in compact form; each is taken up more fully in §2.3. CKS is not RAG; the substrate is not a retrieval index. CKS is not parametric memory; the substrate is external to the LLM. CKS is not a multi-agent protocol; it concerns the human-to-LLM interface over a shared substrate. CKS is not a governance platform; governance content lives inside the substrate, not around the runtime. CKS is not a complete system architecture; it is a design pattern at one layer, composing with others rather than replacing them.

1.3 Roadmap — three-part structure

The paper is organized in three parts. Part 1 articulates and defends the core CKS theory through Claims 1 through 5, covering the substrate as a human-governed coordination artifact, the governance boundary between substrate and LLM, conflict preservation at the substrate level, linear-cost storage and composition, and the tool-agnostic cell-level pattern. Part 2 presents the human-AI-human coordination extension through Claim 6, in which the same pattern is applied to multi-human, multi-role settings. Part 3 presents the proof-of-concept as an integrated demonstration of the core theory and the extension together. Related Work appears at the end of Part 1 for core-theory neighbors and within Part 2 for multi-human work specifically.

The POC demonstrates two axes of benefit, and both are previewed here because the paper’s contribution holds even for solo users, with multi-human coordination as an additional capability the same architecture supports. The individual-author axis is AI-assisted lookup of existing schemas, decision rationale, and prior logic across sessions — the substrate carrying forward, for a single author, what was decided earlier and why. The multi-human axis is cross-role cross-session coordination across business analyst, developer, and reviewer interactions — the same substrate mediating between participants who are not the same person. Full treatment of both is in §12.3.

Part 1 — Core Theory

2. Framing and the Core Design Pattern

2.1 The CKS design pattern — architectural commitments

CKS has two levels of structure, substrate and cell, and the distinction between them is load-bearing. The substrate is the atomic unit: a structured representation of one informational task — entities, relationships, decisions, rationale, and contradictions — typically viewable as a 2D table but not restricted to that form if another shape serves its users better. The cell is the functional whole: a coordinated collection of substrates, possibly just one for a simple task, together with human-authored orchestration rules that determine how the content is used when the cell executes. The substrates hold content; the orchestration rules decide what happens with it. Both levels are human-authorable and human-inspectable: a person can read what is there, write new content, and change the rules that govern cell-level behavior.

The design pattern is described by six architectural commitments, stated here without defense; the defense is the work of Claims 1–5.

1. **Hybrid with a governance boundary.** The substrate handles coordination and governance; the LLM handles high-dimensional reasoning; the division between them is drawn at the governance boundary, not the capability boundary.

2. **Conflict-preserving.** Contradictions, inconsistencies, and unresolved ambiguities are preserved in the substrate by default as first-class addressable objects with their own identity and provenance; cell-level responses to preserved conflicts follow human-authored orchestration rules rather than LLM-driven resolution.
3. **Human-governed, not human-generated.** Humans hold ultimate authority over substrate content, structure, and orchestration rules; content may be originated by humans or by LLMs operating under human direction.
4. **AI as substrate mediator.** The LLM helps the substrate and its users read, write, interpret, and compose substrate content, and executes cell operations in accordance with human-authored orchestration rules.
5. **Tool-agnostic.** Any environment supporting persistent structured state, human read/write access, and LLM access to substrate content can instantiate the pattern.
6. **Linear-cost in storage and composition.** Substrate expansion scales like a database — adding cells or substrates grows storage linearly — without the superlinear costs typical of parameter growth, re-embedding, or agent coordination.

The ordering runs from architectural foundations upward to empirical consequences, and the rest of the paper inherits that flow.

One vocabulary distinction earns its keep throughout the paper, and it is worth making once here explicitly. The architectural commitment behind “human-governed” is a commitment to human *authority*: humans retain the ability to inspect, modify, and override substrate content and orchestration rules at any time. It is not a commitment to human *labor*. Authoring, curation, and maintenance are real work. Humans may perform it directly; LLMs under human governance may perform it on behalf of humans; stable cells may largely automate it under orchestration rules. All three modes are available and widely used. What is not optional is the authority — who holds final say when something has to be decided or changed — and “governed” is the word the paper uses for that authority-layer commitment. Wherever the paper describes human work as work (drafting, curating, maintaining), it means labor in the ordinary sense; wherever the paper says “human-governed,” it means authority.

A second phrase is introduced here because Claim 1 and later sections depend on it: **at the coordination and decision layer**. Data catalogs and enterprise knowledge bases sit at the asset layer — they describe what data exists and what it means, so retrieval and analysis over that data can be done consistently. CKS sits one layer up, at the coordination and decision layer: the substrate captures what has been decided, by whom, under what authority, with what rationale, and where contradictions remain. Whenever the asset-layer reading threatens to absorb CKS — most often when adjacent work from data governance or enterprise knowledge bases is in view — the paper invokes this phrase to pull CKS back to its proper layer.

2.2 Core-vs-extension structure of the theory

The claims are organized to reflect this core-and-extension split. Claims 1 through 5 defend core-theory commitments: the substrate as a human-governed artifact at the coordination and decision layer, conflict preservation as first-class substrate state, the governance boundary between substrate and LLM, linear-cost scaling, and tool-agnosticism. Claim 6 defends the human-AI-human coordination extension, in which the same pattern is applied to multi-human settings and the cell mediates coordination across participants in different roles. The split is deliberate rather than cosmetic. The core theory is self-contained at the single-human scope and can be read, evaluated,

and applied without invoking the extension; the extension builds on the core theory without replacing it. A reader persuaded of Claims 1–5 has a complete pattern to work with, even if Claim 6’s multi-human extension is set aside.

Other extensions — substrate interoperability across organizations, substrate-as-regulatory-artifact, substrate-based institutional memory, and substrate-mediated human-AI-agent coordination — are acknowledged as directions for follow-on research rather than as contributions of this paper, because the core theory plus one defended extension is the scope the paper claims.

2.3 What CKS is not

The five disambiguations from §1.2 each deserve a sentence of explicit handling, because each names a neighboring pattern into which CKS is commonly collapsed. CKS is not RAG: the substrate is a coordination artifact humans read from, write into, and hold authority over, and its primary function is to encode decisions, rationale, and preserved contradictions rather than to retrieve source material. CKS is not parametric memory: the substrate is external to the LLM and deterministic, and its state changes through human-governed writes rather than through training. CKS is not a multi-agent protocol: the pattern concerns the interface between humans and LLM assistance mediated by a shared substrate, not how multiple AI agents coordinate with each other. CKS is not a governance platform: runtime governance middleware wraps the LLM with policies enforced around the edges of the runtime, whereas CKS locates governance content and orchestration rules inside the substrate humans directly govern. CKS is not a complete system architecture: it is a design pattern at the coordination-knowledge layer that composes with reasoning-execution, coordination-mechanism, and control-plane layers rather than replacing them.

One further disambiguation is worth stating once explicitly, because the authority-vs-labor distinction from §2.1 is easy to misread and the misreading is common. CKS is not human-only work. LLMs can draft substrate content, populate new substrates from existing materials, restructure and refactor existing substrates, and in stable cells fully execute substrate content under human-authored orchestration rules — including resolving conflicts when those rules authorize the resolution. What humans own is authority (the right to inspect, modify, and override at any time), format discipline (the commitment to keep substrate structure coherent and auditable), and the orchestration rules that govern how cells behave. Everything else is labor the architecture supports but does not demand.

3. Claim 1 — The substrate as human-governed coordination artifact at the coordination and decision layer

3.1 Claim statement and distinction from adjacent strands

Claim 1 states the paper’s first architectural commitment: effective LLM-era coordination in high-stakes settings calls for treating the knowledge substrate itself — not merely its IO schemas or its retrieval index — as a human-governed structured artifact at the coordination and decision layer, whose primary design goals are traceability, auditability, conflict preservation, and governance. Pieces of this pattern already exist in structured RAG variants, design-rationale knowledge graphs, data-governance catalogs, and the emerging schema-first prompting discourse; CKS articulates them as a unified design pattern applicable across domains, designed for non-specialist governance

(with non-specialist authorship demonstrated by the POC), and instantiated in a working regulated-domain setting. The claim is a core-theory commitment: it applies to the single-human coordination scope introduced in §2.2, and nothing in it depends on multi-human participants being in play.

Four adjacent strands in the 2024–2026 literature each cover a slice of what CKS claims as a unified design pattern. Structured and graph-augmented RAG — GraphRAG (Edge et al. 2024), CogGRAG, CogitoRAG, TaSR-RAG — places graphs or taxonomies in front of retrieval, but the graph is a derived index over a corpus and the primary design goal is retrieval quality rather than governance. Design-rationale and decision knowledge graphs — Kantara (Dhaouadi, Oakes, Famelis 2022) and C-RFBS — treat decisions and rationale as first-class graph content, but scope the pattern to software and engineering design and typically populate the graph by extraction from historical artifacts. Data-governance catalogs and enterprise knowledge bases — Atlan (industry source), Fluree (vendor), Glean’s guidance on graph-versus-vector architectures (vendor), TrustGraph (vendor characterization) — treat the substrate as governed infrastructure with certification, access control, and audit evidence, but operate at the asset layer (this table, this document, this business term) rather than at the coordination and decision layer. Schema-first prompting and structured context engineering — DST-as-prompting, SchemaPro, and Anthropic’s context-engineering essay (practitioner source) — specify IO contracts between humans, code, and LLMs and prescribe how context is packed, but do not commit to any particular shape for the underlying knowledge artifact. Each strand names a real slice of the territory; no strand asks the substrate to be all four things at once — human-governed in origin, decision-and-rationale in content, governance-constitutive in design goals, and coordination-layer in scope.

Among the four strands, Kantara (Dhaouadi, Oakes, Famelis 2022) is the closest single neighbor and deserves sharpened differentiation. Kantara is an end-to-end pipeline that uses NLP and information extraction to reconstruct a knowledge graph of decisions and their rationale from software-engineering artifacts — commits, issues, pull requests — and treats that graph as authoritative substrate for downstream AI assistance. The shared commitments are substantial: rationale and decision provenance as first-class graph content, the KG as the artifact downstream AI operates over, traceability as a constitutive design goal. The load-bearing difference is where structure originates. Kantara extracts structure from prior artifacts; CKS requires the substrate to be human-governed at the coordination and decision layer, with structure originating under human authorship and automated extraction admissible only as drafting assistance operating under that governance. This paper names the axis “extraction-from-prior-artifacts versus authoring-under-human-governance,” and §3.2 defends it.

A separate cross-cutting thread runs through the academic literature on KG-backed AI accountability and explainability, and it matches CKS’s design goals — traceability, audit, provenance — more directly than any of the four strands above, even as its scope typically narrows to a single technology stack. Naja, Markovic, Edwards, and Cottrill (2021) use PROV-O and EP-Plan to model governance as queryable graph structure, distinguishing an accountability plan — what the system specifies must be captured across the AI lifecycle — from an accountability trace — what actually occurred in any given run. Baldazzi, Bellomarini, Ceri and colleagues (2024) combine a rule-augmented enterprise KG with LLM verbalization in KGLM, producing deterministic provenance-path explanations that an LLM renders in natural language. Zhang, Meroño-Peñuela, and Simperl (2023) argue that KG lifecycle transparency and accountability remain limited and that the substrate itself, not only what rides on top of it, needs governance as AI regulation matures. Phiri (2025) makes the complementary case at the agent layer: without explicit recording, agentic systems have no decision trail. Rajabi and Kafaie (2022) offer the formulation the paper reuses

for the mechanism of decision traceability — path retraceability over explicit structure. These are the vocabulary imports the paper commits to once, here: accountability plan, accountability trace, and path retraceability. They do work across Claims 2 through 6 without further definition. What the thread does not do — and what CKS does — is unify these commitments into a single design pattern that treats the substrate as a coordination and decision artifact applicable across domains and across technology stacks, rather than instantiated in a single RDF / OWL / SHACL pipeline.

3.2 Extraction vs. curation as the load-bearing axis

The load-bearing axis is the origin of substrate structure. CKS’s primary origin is authoring under human governance; Kantara’s primary origin is extraction from prior artifacts. The paper states the axis as a primary-origin claim rather than an absolute one, because neither pattern is pure. CKS permits LLM-drafted substrate content in every architectural commitment the core theory states, so long as humans retain authority over that content; Kantara may incorporate human review downstream of extraction. What separates the two works is which end of the pipeline originates the structure that goes on to be treated as authoritative.

The axis matters because in regulated coordination the substrate is not merely input to downstream reasoning; it is the governance artifact the organization stands behind when audited. An artifact extracted from prior text inherits its content from that text, whatever its provenance and whatever its errors; it does not certify what it contains. An artifact authored under governance is the certification — the thing a reviewer, an auditor, or a regulator can point to as the record of what was decided, by whom, under what authority, with what rationale. Mushtaq, Trovati, and Bessis (2025) provide a useful peer-reviewed precedent for this stance in the LLM era: an expert-curated oncology knowledge graph grounded in PubMed, trial reports, and guideline publications, backing an LLM-era retrieval system in a regulated domain. Theirs is one of the few published instances in which a substrate’s governance weight comes from expert curation rather than from extraction, and it confirms that the distinction matters in practice at the coordination and decision layer, where the cost of audit failure is high.

One honest qualification belongs here. The evidence base for Kantara available to this paper consists of summaries and secondary characterizations that foreground the extraction pipeline; full-text reading may yet reveal a richer human-curation step elided by those summaries. If that is the case, the axis framing tightens rather than reverses: “authoring-under-governance versus extraction-with-downstream-review” remains a real distinction on the governance-weight dimension, because downstream review is a different commitment than origin-level authorship in the same way that signing off on a draft is different from writing it. The paper states the axis as the evidence supports it — Kantara’s primary origin is extraction, CKS’s primary origin is authoring under governance — and flags the qualification here rather than letting it surface later as a reviewer objection.

3.3 Kantara reviewer-attack response

One reviewer objection anticipates itself at this point, and the paper’s response takes the form of a phrase it deploys in every section where the objection surfaces: **Governance is an authority architecture, not a review workflow.** The claim’s use of “human-governed” names a commitment about where authority sits, not about whether a human reviews every element. Authority sits in the human-governed substrate, in the human-authored orchestration rules that determine cell-level behavior, and in humans’ preserved right to override at any moment. Review may or may not occur at any given element; the commitment the architecture makes is that authority

remains available wherever review is chosen to happen. The authority-vs-labor distinction this paper introduced in §2.1 is what the phrase renders portable; §7.4 extends the same distinction to non-specialist governance when Claim 5 takes up tool-agnosticism.

The specific misreading the phrase preempts runs as follows. A reviewer reading Claim 1 may take “human-governed substrate” to imply that a human reviews every element the substrate contains — every cell, every rule, every conflict, every LLM-drafted addition — and may conclude, reasonably on that reading, that CKS scales poorly because human review is a bottleneck that grows with substrate size. The reading collapses CKS into human-in-the-loop review, reviewer-in-the-loop AI governance, or Kantara-style extraction-with-downstream-human-sign-off, all of which are distinct patterns the paper does not claim to be. The response is concrete: authority in CKS is exercised at two moments, neither of which is a per-operation review checkpoint. Authority is exercised when humans write the orchestration rules that govern cell-level behavior — who can modify the substrate, how conflicts are handled, what the LLM is authorized to do on the substrate’s behalf — and when humans use their preserved override right to directly modify substrate content. The first cost is paid once at design time and amortizes across every cell execution that follows; the second is paid only when a human chooses to intervene. Neither cost is proportional to substrate size. The review workflow a reviewer might infer from “human-governed” — a human approving every substrate change in series — is not the architecture this paper defends, and is not what the word “governed” commits CKS to.

4. Claim 2 — The governance boundary and schema-level decomposition

4.1 Claim statement and the “what is schematized” argument

Claim 2 states the governance boundary as the architectural commitment that specifies the substrate/LLM division of labor, and the schema-level decomposition that makes the substrate coherent as a design object. CKS draws the substrate/LLM division of labor at the governance boundary, not the capability boundary: the substrate handles coordination and governance (deterministic, human-governed, auditable); the LLM handles high-dimensional reasoning that humans cannot or should not encode. In this division, the LLM operates as mediator over the substrate — helping humans read, write, interpret, and compose substrate content, and executing cell operations in accordance with human-authored orchestration rules — rather than as an autonomous agent over the workflow. The substrate is always the authoritative artifact; the LLM operates relative to it. This AI-as-substrate-mediator role is a core-theory commitment that applies to single-human coordination and extends to multi-human coordination in Claim 6. CKS decomposes the substrate into small, scoped cells with narrow decision boundaries through schema-level structure — specifically schematizing domain knowledge rather than the agent-level role assignment typical of multi-agent coordination frameworks (Chang; mainstream MAS).

Four architectural layers are visible across the 2024–2026 literature that most directly shares CKS’s concerns. Reasoning execution: Seck’s Cognitive Core (2026), which decomposes reasoning into typed epistemic primitives (retrieve, classify, verify, deliberate, and so on), each with its own governance profile. Coordination mechanisms: Chang’s MACI (2026), which proposes baiting, filtering, and persistence as the primitives binding multi-agent pattern outputs into reliable behavior. Control plane: Stevens’s work at Sakura Sky (2025) and the adjacent human-in-the-loop governance

literature, which treats approval gates, reviewer workflows, and accountable decision logs as the primitives enforcing human authority over autonomous action. Coordination knowledge: the layer where CKS sits, in which the substrate carries human-governed domain content — descriptions, sample values, logic, decision rationale, authority policies — that humans and models both read from and write into. §4.3 presents these four layers as a single diagram; §4.4 differentiates CKS from the three neighbors on their own layers’ terms.

The load-bearing move at this layer is the content-versus-operations distinction. Cognitive Core schematizes reasoning execution: what are the typed epistemic operations a system performs under governance, and how does each operation’s governance profile determine when it is applied. CKS schematizes coordination knowledge: what is the human-governed domain content a workflow coordinates around — the business descriptions, the sample values, the SQL logic, the decision rationale, the authority policies — and how does that content carry its own provenance, conflict state, and authorship. The two schemas answer different questions about what gets decomposed. They are complementary layers: CKS cells are candidate inputs to Cognitive Core primitives, not substitutes for them. The paper returns to this framing in §4.4 as the sharpest axis of differentiation between CKS and its closest named neighbor.

4.2 AI-as-substrate-mediator as core-theory commitment

The AI-as-substrate-mediator role Claim 2’s statement names deserves subsection treatment of its own, because its architectural consequences reach beyond the immediate governance boundary into the cross-claim structure of the theory. The commitment applies to the single-human coordination scope Claim 1 established — even when one person is coordinating structured work across sessions, the substrate holds authority over what has been decided, and the LLM’s reads, writes, and cell executions happen relative to that substrate rather than in place of it — and the same commitment carries into the multi-human coordination Claim 6 develops. This subsection states the commitment once, as core theory, so Claim 6 can extend rather than re-assert it.

The mediator commitment is also the cross-claim spine that connects Claim 2 to Claim 6, and stating the spine once is the work of this paragraph. When the core theory is instantiated with multiple humans in different roles, the cell — aggregating substrates and LLMs operating under this same mediator commitment — comes to function as a mediator of coordination across participants. Claim 6 defends that extension. What matters here is that Claim 6 does not re-introduce the mediator role from first principles; it carries this core-theory commitment into the multi-human context as its architectural foundation. A reader persuaded of Claim 2’s mediator commitment and of the substrate’s persistence across sessions has already accepted the architectural move Claim 6 develops; Claim 6’s additional work is to defend the coordination-across-participants property as an emergent consequence of the core pattern rather than as a new commitment.

4.3 Four-layer landscape diagram

The four layers §4.1 named can be rendered as a single architectural diagram, and the paper uses that diagram in this section as its anchor for the cross-neighbor comparisons that follow. The diagram arranges the four layers — reasoning execution, coordination mechanisms, control plane, and coordination knowledge — as a stack of architectural concerns visible in the 2024–2026 governance-first literature, with a representative work occupying each layer. Cognitive Core (Seck 2026) anchors the reasoning-execution layer as the clearest instance of schematizing typed epistemic operations under governance. MACI (Chang 2026) anchors the coordination-mechanisms layer as

the clearest instance of schematizing multi-agent coordination mechanisms rather than content. Stevens and Sakura Sky (2025) anchor the control-plane layer as the clearest instance of governance primitives sitting around whatever reasoning the agent is doing. CKS anchors the coordination-knowledge layer as the instance this paper defends. The diagram’s work is analytical, not rhetorical: it separates four architectural moves that have been collapsed in the landscape discourse, and it locates the four representative works on their own layers rather than on competing diagonals.

The diagram lives here in §4.3, in the body of Claim 2, rather than in Related Work. It does load-bearing work for the architectural-layer argument and belongs where that argument is made. Claim 5’s tool-landscape diagram — a four-quadrant market map covering execution substrates, platform governance, citizen-developer tooling, and shadow AI — lives in §7.2 for the same reason: it anchors Claim 5’s empty-region argument where that argument is made. The paper uses two diagrams rather than a unified one because the two answer categorically different questions — this one is an architectural taxonomy of governance-first concerns; §7.2’s is a market map of 2024–2026 tool affordances — and forcing a unified rendering would compromise both.

4.4 Complementarity with Cognitive Core; differentiation from MACI and control-plane work

Cognitive Core is the closest named neighbor, and the sharpest thing this paper can say about the relationship is that CKS and Cognitive Core are different design objects, not different layers of the same architectural stack. Seck’s paper (2026) develops Cognitive Core as a design methodology for deploying AI systems in regulated coordination contexts; its primary commitments are agent-orchestration patterns captured in a three-layer configuration model — workflow YAML (the epistemic sequence a system performs), domain YAML (the domain expertise the sequence draws on), case JSON (the runtime data of a particular instance). Those orchestration patterns apply across underlying AI approaches: Cognitive Core’s architecture is deliberately compatible with traditional rule-based systems, fine-tuned models, and LLM-assisted workflows. In the vocabulary §4.1 introduced, Cognitive Core is substrate-agnostic. CKS is something else. It is a design pattern for the substrate itself — what the substrate is as an architectural object, what properties it carries, how governance lives inside it. The two works address different design objects, and the consequences of that difference run in both directions. A Cognitive Core deployment could use a CKS substrate as its domain layer, but Cognitive Core can equally use any other substrate that satisfies its configuration requirements, including substrates that make none of the commitments Claim 1 defends. A CKS instantiation can sit inside a Cognitive Core deployment as its domain content, but CKS can equally be instantiated without Cognitive Core’s agent-orchestration methodology, as the POC demonstrates. The honest framing is “different design objects” rather than “complementary layers”: neither work is making a claim that subsumes or is subsumed by the other, and neither needs the other to be correct. One further observation sharpens this reading. Cognitive Core’s text uses the word “substrate” — Seck calls the Cognitive Core itself a governed decision substrate — but Cognitive Core does not treat the substrate as a design object in its own right. It treats substrate content as configuration material for the agent-orchestration model the paper designs. CKS’s architectural move is exactly what Cognitive Core does not make: treating the substrate as a design object whose properties must be defended, rather than as configuration material whose properties can be left to the instance.

MACI (Chang 2026) is the same-diagnosis-different-architecture neighbor. Chang argues that the AGI research program has stalled at pattern repositories — LLMs trained on vast distributions but unable to translate competence into reliably coordinated behavior — and names a missing coordi-

nation layer above those repositories as the architectural fix. That diagnosis rhymes with the gap CKS identifies in structured work. Where the two works separate is what the coordination layer contains. MACI proposes coordination mechanisms — baiting (debate patterns that modulate agent behavior), filtering (Socratic judging via criteria-based gates), persistence (transactional memory across steps of a reasoning session) — and formalizes them through Unified Contextual Control Theory. CKS proposes coordination knowledge: human-governed domain content that participants coordinate around. The two layers are different objects at different scopes. MACI is AGI-oriented and AI-coordinated; its coordination happens among agents through mechanisms the system imposes on agent interactions. CKS is institutional-coordination-oriented and human-governed; its coordination happens among humans, and between humans and LLMs, through content a human authority stands behind. Chang’s scope is at the agent-interaction level; CKS’s scope is at the schema level. The missing-coordination-layer framing is shared vocabulary; the layer each paper fills is not.

Stevens and Sakura Sky’s “Missing Primitives for Trustworthy AI Agents” series (2025) names the control-plane primitives that institutional AI deployments need — approval gates, reviewer workflows (triage, review, annotate, approve or reject, escalate, log), break-glass override, accountable decision logs — and positions them as the architecture that transforms agents into governable systems aligned with organizational responsibility. These primitives sit around whatever reasoning the agent performs; they enforce human authority at workflow breakpoints and record what crossed which gate. CKS sits underneath, as the substrate content the reasoning draws from and writes into. The two works compose cleanly: approval gates around a cell execution are Stevens’s primitive; the cell’s content, orchestration rules, and substrate state are CKS’s design object. Stevens complements CKS rather than competes with it, and the combination is stronger than either work alone — but because the layers are different, neither work subsumes the other, and a CKS deployment need not adopt Stevens’s specific control-plane primitives to be CKS.

5. Claim 3 — Conflicts as first-class addressable objects

5.1 Claim statement and the triangulating trio

Claim 3 states the paper’s architectural commitment about what the substrate carries at the conflict layer: contradictions, inconsistencies, and unresolved ambiguities encountered during AI-mediated coordination are persisted in the substrate as first-class, addressable structured objects with their own identity and provenance — attached directly to the conflict itself (which rule, from which prior decision, under which role or context), not only to downstream decisions the conflict influenced. These objects are re-addressable across sessions, so the same conflict can be looked up, updated, or reactivated in a future workflow rather than merely reconstructed from logs. At the substrate level, conflicts are preserved by default; cell-level responses to preserved conflicts follow human-authored orchestration rules rather than LLM-driven resolution. Human deferral — explicitly choosing not to resolve a conflict at the moment it surfaces — is a terminal recorded outcome for such objects rather than a failure mode. This extends a well-established stance from paraconsistent logic and computational argumentation — that contradictions can be rational, persistent elements of a knowledge state rather than anomalies requiring immediate repair — into the design of LLM-mediated coordination substrates, and contrasts with the detect-resolve-forget pattern prevalent in current agent frameworks — contradictions detected at inference time, reconciled in-session through clarification, majority voting, or judge-LLM arbitration, and discarded once the session ends.

The 2019–2026 literature contains three closest neighbors for Claim 3, each of which supplies one of the claim’s load-bearing architectural ingredients. Chapman, Kökciyan and colleagues’ CONSULT (2019) is the closest single neighbor on conflict-as-object-with-identity and argument-level provenance: CONSULT is an argumentation-based clinical decision support system whose pro and contra arguments for treatment options are constructed from EHR, sensor, and guideline sources, weighted by source quality, and preserved for clinician review. Provenance attaches to the argument itself rather than to the downstream treatment decision, and human review is the terminal interaction. What CONSULT does not commit to — in the evidence base available to this paper — is cross-session re-addressability of the conflict objects as a first-class design primitive; it is implied by EHR integration rather than named as an architectural property, and the scope is clinical decision support rather than general coordination. Jin, Moon, and Žitnik’s StratCP (2026) is the closest single neighbor on deferral-as-typed-outcome: a stratified conformal framework for medical foundation models that partitions every case into an action arm (FDR-controlled decision) or a deferral arm (calibrated prediction set with coverage guarantee), structurally distinguishing deferral from decision and returning a decision-ready object for the deferred case. What StratCP does not do is attach deferral to a conflict-as-object with its own identity; the deferred entity is the case, provenance is implicit in the model and calibration process rather than attached to the deferral output, and the scope is single-encounter inference. Allen and colleagues’ “Sound and Complete Neuro-symbolic Reasoning with LLMs” (2025) is the closest single neighbor on formal logical grounding: Belnap four-valued (paraconsistent) semantics integrated into an interpretation function where the LLM is the knowledge source and “Both” — contradicted — is a valid semantic assignment rather than an error. What Allen et al. do not provide is a coordination substrate; the contribution lives at inference-time formal semantics, with no cross-session mechanism and no human-facing representation of conflict state. The synthesis observation that matters is that the four load-bearing elements of Claim 3 — conflict-as-object with identity, provenance attached to the conflict, deferral as terminal recorded outcome, and cross-session re-addressability — each appear in one of these works, but no single work carries all four as architectural commitments of a general coordination substrate.

The philosophical lineage the claim imports deserves explicit acknowledgment, because Claim 3’s contribution is the architectural operationalization of a stance rather than the founding of one. The stance that contradictions can be rational, persistent elements of a knowledge state rather than anomalies requiring immediate repair is the inheritance of paraconsistent logic — consequence relations that are not explosive — and of the Logics of Formal Inconsistency tradition, in which consistency and inconsistency are internalized as first-class notions inside the object language rather than treated as metatheoretic properties the system must restore. The structural vocabulary for treating pro and contra positions as addressable graph content is the inheritance of Dung-style abstract argumentation, in which arguments are nodes, attack and support are edges, and multiple candidate stances over inconsistent information coexist as extensions. CKS does not extend either tradition in its own terms — it does not introduce new paraconsistent semantics, and it does not propose new argumentation frameworks or extension semantics. What Claim 3 contributes is the operationalization of this stance as a general coordination substrate: contradictions live in a persistent, human-governed artifact outside the LLM, carry provenance attached to the conflict itself, remain re-addressable across sessions, and are acted on at cell execution time through human-authored orchestration rules. The lineage supplies philosophical warrant; the architectural work is CKS’s own.

5.2 OIDA insertion — signed contradiction edges as adjacent precedent

A distinct line of 2026 work, more recent than the triangulating trio and architecturally closer to CKS’s substrate-level move, warrants its own treatment here. OIDA — “Retrieval Is Not Enough: Why Organizational AI Needs Epistemic Structure” (2026 preprint, not peer-reviewed at time of writing) — structures organizational knowledge as typed Knowledge Objects that carry an epistemic class (fact, hypothesis, decision, or question), an importance score with class-specific decay, and *signed contradiction edges* between conflicting objects; a deterministic Knowledge Gravity Engine maintains the substrate over time, and QUESTION-as-modeled-ignorance is treated as a first-class primitive rather than as a pending-resolution state. Zahn and Chana’s Knowledge Objects paper (2026) occupies the same architectural family — typed, structured knowledge units as first-class artifacts — and supplies the precedent vocabulary OIDA builds on. The shared commitment across OIDA and Knowledge Objects is that coordination-grade knowledge is a schema-level object with typed structure and maintained state, not a retrieval target recovered from a corpus at inference time.

OIDA matters to Claim 3 because its signed contradiction edges and its QUESTION-as-modeled-ignorance primitive are the strongest 2024–2026 precedent for treating contradictions and missing knowledge as first-class, addressable substrate state — precisely the architectural move Claim 3 defends. A signed contradiction edge between two Knowledge Objects is a structural realization of conflict-as-object with identity: the edge has its own schema presence, it points to both conflicting objects, and it persists in the substrate rather than existing as transient state consumed in a single inference pass. Modeled ignorance — the commitment that what a system does not know is substrate content in its own right, not the absence of substrate content — is the analogue, on the missing-knowledge axis, of what Claim 3 commits to on the contradiction axis. OIDA’s presence in the 2026 literature does not scoop Claim 3; it validates the design pattern. Adjacent prior art on the contradiction-representation axis strengthens Claim 3’s defensibility rather than weakening it, because the claim’s contribution is not the novelty of the representational move but the architectural combination Claim 3 names — substrate-level preservation, provenance attached to the conflict itself, re-addressability across sessions, deferral as terminal outcome, and all of this under the governance commitments Claim 1 and Claim 2 defend.

CKS differs from OIDA on three architectural axes, and stating them precisely matters because the adjacency invites exactly one reviewer misreading: that CKS is OIDA extended to multi-human use. The first axis is governance. CKS commits to human governance of the substrate as a design requirement — the substrate is human-governed in origin and under preserved human authority for modification and override at any time — whereas OIDA does not specify a governance architecture; it specifies a maintenance architecture, the Knowledge Gravity Engine, which operates over the substrate deterministically but does not locate authority over substrate content in any particular actor. The second axis is role and authority. CKS encodes formal role and authority semantics at the cell level — who may decide what, on what evidence, under which policy — as substrate content the cell’s orchestration rules draw on; OIDA does not commit to a role/authority schema, and its Knowledge Objects carry epistemic class but not institutional provenance in the role-authority sense. The third axis is the AI’s architectural role. CKS frames AI as mediator over the substrate, helping humans read, write, interpret, and compose substrate content and executing cell operations under human-authored orchestration rules; OIDA frames AI as substrate client, consuming structured memory to answer queries rather than operating as mediator over the substrate’s human-facing state. These three axes are not gradations of the same commitment; they are independent architectural moves that together characterize a different design posture than the

one OIDA takes. CKS is **architecturally different from OIDA on the multi-human axis, not OIDA plus multi-human.**

5.3 Two-level conflict handling

Conflict handling in CKS operates at two levels, and the relationship between them is what keeps Claim 3 from collapsing into either a CKS-never-resolves-conflicts commitment or a CKS-resolves-conflicts-the-way-everyone-else-does commitment. At the substrate level, conflicts are preserved by default: the substrate does not auto-resolve contradictions, and no LLM operation over the substrate may silently merge, discard, or force resolution of them. At the cell level, when a cell executes and encounters a preserved substrate-level conflict, the cell’s response is determined by orchestration rules humans authored in advance — rules that may direct the cell to preserve and skip, to surface the conflict to a participant, to delegate it to another substrate designed for adjudication, to defer with a recorded outcome, or to follow a pre-specified resolution logic under specified conditions. Resolution is therefore available in CKS, not absent from it. Humans exercising authority over the substrate can always modify substrate content directly — rewriting rules for clarity, adding superseding rules, or recording which rule wins — and humans authoring orchestration rules can specify conditions under which cell execution resolves conflicts under delegated authority. The architectural constraint Claim 3 commits to is not on *whether* conflicts get resolved; it is on *who decides* resolution logic. Humans decide, either directly by editing the substrate or indirectly by authoring the rules that govern cell execution. The LLM does not.

The two-level structure produces a clean behavioral distinction the rest of the paper relies on. Substrate-level preservation is an architectural default: the substrate’s schema treats conflicts as first-class content whose presence cannot be erased by any LLM operation authorized under any delegation level. Cell-level response is a question of executed behavior: it is determined by human-authored orchestration rules that specify what should happen when a cell execution encounters a preserved conflict under given conditions. In stable cells — those where humans have established sufficient trust through iteration to delegate substrate modification to the LLM — the LLM executes the orchestration rules as written, may modify substrate content in ways those rules authorize (including conflict-resolving modifications the rules specify), and preserves any conflict the rules did not address. What the LLM cannot do, under any delegation level, is resolve conflicts outside what the rules authorize, or override the substrate’s conflict-preservation default for conflicts the rules did not address. This separation is what lets CKS scale: humans exercise authority at two tractable points — when they author the orchestration rules, and when they edit substrate content directly — rather than at every conflict moment during runtime. This section carries the full two-level treatment; Claims 3 and 6 reference the governance boundary and the multi-human coordination extension without restating the substrate-level-versus-cell-level distinction.

5.4 Detect-resolve-forget as the named foil

The dominant alternative pattern in current LLM agent frameworks and in adjacent contradiction-handling research has a handle worth naming: **detect-resolve-forget**. Contradictions are detected at inference time, reconciled in-session through clarification, majority voting, or judge-LLM arbitration, and discarded once the session ends. The 2024–2025 literature instantiates this pattern in several concrete lines. RAG contradiction detection paired with a context validator treats conflicts in retrieved documents as a retrieval-quality problem: an LLM judge classifies and segments the conflicts, and a filter produces a consistent context before generation. “Explanation Generation for Contradiction Reconciliation with LLMs” formalizes success as flipping a label from contradiction

to entailment through the generation of additional explanatory context — a move whose architectural premise is that the contradiction existed only because the system lacked enough context to dissolve it. Clarification-question-generation agents — SAGE-Agent, “Active Task Disambiguation with LLMs,” factual clarification-question benchmarks, and preference-elicitation funnels — treat ambiguity as a control-policy problem, optimizing clarification-question selection by expected information gain or Bayesian experimental design, and overwriting prior user-profile state with refined estimates once the clarification is collected. LegalWiz treats structured contradictions as test inputs for stress-testing RAG systems, inverting the representational direction: contradictions are synthesized to evaluate a detection-and-resolution capability rather than preserved as live substrate state. Across these lines, the commitment is the same: contradictions are transient, in-session, and reconciled before the system emits.

Claim 3’s contrast with detect-resolve-forget is architectural in nature, not efficacy-based. The two patterns address the same surface phenomenon — ambiguity encountered during AI-mediated work — with opposite stances. Detect-resolve-forget collapses the space of live alternatives before emission, treating contradiction as a transient state whose resolution is the system’s job. CKS preserves the space of live alternatives in the substrate, treating contradiction as substrate content whose addressability across sessions is part of the architecture. The stances serve different design goals. Clarification-question agents have legitimate uses; they are the right pattern for single-turn information-gathering tasks where closing the loop before proceeding is the design intent, and overwriting prior state with refined estimates is an accepted cost. CKS is the right pattern for coordination settings where the reasoning trail behind *what was decided, by whom, under what authority, with what unresolved contradictions* matters as much as the decisions themselves, and where collapsing that trail at session boundaries forfeits exactly what the coordination artifact exists to preserve. Neither pattern is better in the abstract; each is better at the design goal its architecture is optimized for. What the paper commits to is that substrate-level preservation is the right architectural stance for the coordination settings Claim 1 named and the POC exercises, not the stronger claim that detect-resolve-forget is wrong for every setting.

6. Claim 4 — Linear-cost storage and composition

6.1 Three sub-claims

Claim 4 states the paper’s cost-model commitment: adding coordination knowledge to a CKS system incurs linear runtime/compute cost in the substrate — the cost model is database-like addition rather than parameter growth, re-embedding, or agent coordination overhead. Composition of knowledge across cells is human-governed and selective, not automatic aggregation. This is a claim about infrastructure scaling, not coordination intelligence scaling: CKS does not claim that organizational understanding grows linearly; it claims the underlying substrate does not exhibit the superlinear scaling costs characteristic of fine-tuning, continual learning, knowledge editing, re-embedding, or multi-agent coordination at scale. Authoring and curation cost — the labor of writing substrate content, whether performed by humans directly or delegated to LLMs operating under human governance — is deliberately significant and is not part of the linear-cost claim; it is labor humans choose where to spend, not a mandatory human commitment the architecture imposes. CKS inherits the database-like cost model empirically validated by recent external-substrate work — most directly Zahn & Chana’s Knowledge Objects (2026), which demonstrates $O(1)$ retrieval at $\sim 252\times$ lower per-query cost than in-context memory with cost differences that “grow linearly

as the corpus expands.” CKS extends this architecture with richer governance semantics (beyond the single provenance field KO commits to), human governance commitments, and multi-human coordination primitives. The distinctive framing contribution is treating human-governed selective composition as a scaling mechanism, not merely an accuracy or governance one — a synthesis not explicit in current literature. The framing that earns load-bearing status here is inheritance, not demonstration: CKS commits to the design constraints that produce database-like scaling and cites Knowledge Objects as the empirical anchor for what those constraints yield, rather than claiming to independently validate the cost curve itself.

Claim 4 decomposes into three sub-claims, and the decomposition matters because the three carry different evidentiary weights. Sub-claim 4a is that in-weight and agent-coordination alternatives — fine-tuning, continual learning, knowledge editing, and multi-agent coordination at scale — scale superlinearly. Sub-claim 4b is that database-backed external substrates scale linearly. Sub-claim 4c is that human-governed selective composition functions as a scaling mechanism, not only an accuracy or governance one. Sub-claims 4a and 4b carry quantitative weight from peer-reviewed 2024–2026 work: the cost curves of in-weight accumulation and agent coordination, and the linear-cost curve of database-backed retrieval, are defended in the literature CKS cites. Sub-claim 4c is a synthesis contribution: the ingredients exist across adjacent literatures — algorithmic selective composition, human-curated retrieval, human oversight as governance mechanism — but no 2024–2026 work surfaced by the evidence base directly reframes human selection as a cost lever, and CKS defends the reframing rather than restating an existing result.

6.2 Knowledge Objects inheritance and extension framing

Sub-claim 4a stands on a cross-section of 2024–2026 evidence that is consistent across research, mechanistic, and enterprise-cost voices. Wang et al.’s 2026 survey of continual learning in large language models frames catastrophic forgetting as an unsolved problem at the scales and granularities organizational knowledge updates require, and reports that full retraining and continual pretraining remain computationally prohibitive while still vulnerable to forgetting. A 2026 mechanistic study of catastrophic forgetting grounds this at the parameter level: sequential fine-tuning severely disrupts 15–23% of attention heads in lower layers and drops CKA representation similarity by 0.32–0.47 in intermediate layers, indicating that weight updates are not the additive operation the continual-learning framing invites. Knowledge editing — ROME, MEMIT, and MEND — shows the same non-compositionality at the scaling ceiling: ROME begins catastrophic forgetting between 100 and 1000 edits depending on settings, and MEND degrades toward zero F1 before 100. These are not linear regimes that happen to get expensive; they are regimes in which the cost curve bends at small edit counts. The multi-agent coordination alternative is characterized in the Google-affiliated *Towards a Science of Scaling Agent Systems* (2025–2026), which quantifies coordination overhead growing with interaction depth, agent count, and message density across five canonical architectures, and identifies negative marginal returns beyond a capability ceiling. Atlan’s 2026 enterprise guide (industry source) anchors the cost curves in dollars: 50K–500K USD per full fine-tuning run at the high end, LoRA approximately 10% of that, and a telecom case reporting 2.3M USD in annual savings after switching from periodic fine-tuning to a retrieval-and-knowledge-base architecture. The cost-curve shape is consistent across the evidence: in-weight and agent-coordination accumulation is superlinear and frontier-bending, not linear-in-additions.

Sub-claim 4b has a methodologically rigorous empirical anchor in Zahn & Chana’s Knowledge Objects (2026). KO reports \$0.002 per query versus \$0.57 for in-context memory at N=7,000 — a $\sim 252\times$ cost difference that, in the paper’s language, “grow[s] linearly as the corpus expands” — with

$O(1)$ retrieval via hash lookup and a per-query token cost that holds approximately constant from $N=100$ through $N=100,000$. KO also names the architectural commitment that produces the curve in terms CKS adopts: “separation of concerns between storage and processing,” with facts living in addressable external storage where they cannot decay through compression or be lost at session boundaries, while the LLM retains query understanding, answer generation, and the extraction of new content from ongoing interaction. KO’s “context rot” vocabulary names the failure mode of the in-context alternative — capacity overflow, compaction loss, goal drift under repeated compression passes — that separation of concerns avoids. Adjacent work supports the pattern across voices. Karpathy’s LLM Wiki pattern (2026, practitioner source) endorses plain-text files organized as a wiki and accessed by simple search in place of vector retrieval, with reported efficiency advantages for some use cases. Oracle’s 2026 developer article on files versus databases for agent memory (vendor voice) concludes that databases are preferable for structured, high-cardinality, query-heavy memories due to indexed access, concurrency, and predictable scaling. A 2026 industry guide on knowledge base versus knowledge graph backends for LLM systems recommends KB/KG as the authoritative source of truth with the LLM as a natural-language frontend and structured lookups for fact retrieval. The cost-curve shape these sources describe — linear in storage, $O(1)$ or $O(\log N)$ in lookup, near-constant in per-query tokens — is the regime CKS commits to by design, inheriting the empirical backbone KO supplies.

Knowledge Objects is the closest empirical precedent for sub-claim 4b, and the sharpest thing this paper can say about the relationship is that CKS inherits from KO and extends it on two axes. CKS inherits two commitments. It inherits the database-like cost model validated at $\sim 252\times$ at $N=7,000$, per the previous paragraph. And it inherits the separation-of-concerns architecture between external storage and LLM processing — the commitment that facts live in addressable external storage where they cannot decay through compression or be lost at session boundaries, while the LLM retains query understanding, answer generation, and new-content extraction. On top of these inherited commitments, CKS extends KO on two axes. *First axis — governance semantics.* KO commits to a single `provenance_metadata` field as its governance hook, adequate for attribution and audit at the retrieval-unit level; CKS encodes role and authority, approval states, decision rationale, and conflict state as first-class substrate content, per Claim 1 at §3 and Claim 3 at §5. *Second axis — where disambiguation work lives.* KO disambiguates computationally at retrieval time, via density-adaptive retrieval with learned thresholds (≈ 0.85 in KO’s experiments) and exact structured key matching on (subject, predicate) tuples as fallback; CKS disambiguates through human-authored schema at cell level, making the distinctions humans draw into substrate content encoded at authoring time rather than recovered by runtime algorithm. The two design stances reflect different philosophies about what the substrate is responsible for carrying. KO treats the substrate as storage over which algorithmic disambiguation operates; CKS treats the substrate as the authored artifact whose structure already carries the distinctions. Both senses of “CKS is foundational relative to KO” apply — architectural foundation, because CKS designs what the substrate is as an object, including its schema for role, authority, and rationale content; and philosophical foundation, because CKS takes a different stance about where disambiguation work sits, explicit in authored structure rather than implicit in retrieval algorithm. Sub-claim 4c’s human-governed selective composition fits naturally within this two-axis extension: humans are a selection mechanism operating on the same combinatorial substrate-composition problem KO addresses algorithmically, complementing KO’s algorithmic selection rather than replacing it. Adjacent work on the selection-mechanism dimension sits close enough to name. SciCo-Radar (2024) acknowledges combinatorial blow-up in cross-document reasoning but uses algorithmic selection as the response. Atlan’s data-catalog design has human stewards pre-structure the composition space but frames

the work as precision and governance, not as cost control. FINOS’s AI governance guidance (industry/consortium source) treats LLM-as-Judge and human oversight as risk and safety controls rather than as scaling mechanisms. No 2024–2026 work surfaced by the evidence base directly argues that human selection functions as a scaling lever; CKS defends the reframing rather than restating an existing result. The two-axis extension framing also preempts the reviewer misreading that “CKS is KO with extra fields”: the extra fields are not decoration, they are substrate content doing disambiguation work CKS’s design philosophy requires to be explicit rather than computed.

6.3 Authoring/curation cost as labor humans choose where to spend

One vocabulary distinction earns a third application in §6, and the cleanest place to apply it is here, because Claim 4’s cost-model scope and the authoring and curation labor the architecture supports are not the same thing. The authority-vs-labor distinction §2.1 introduced and §3.3 rendered portable as **Governance is an authority architecture, not a review workflow** applies directly to Claim 4: authoring and curation cost — the labor of writing substrate content, whether performed by humans directly or delegated to LLMs operating under human governance — is deliberately significant and is not part of the linear-cost claim. The linear-cost claim is about infrastructure scaling: adding a cell or a substrate grows storage linearly, and composition stays human-selective rather than exhaustive. The authoring and curation cost is a separate quantity — it is labor humans choose where to spend, not a mandatory human commitment the architecture imposes. CKS asks for authority over substrate content and orchestration rules; it does not ask humans to write every cell.

A specific critique the distinction preempts runs as follows: but humans writing substrate content is expensive, so CKS’s cost model is overstated. The architecture does not claim otherwise. The linear-cost claim is about infrastructure — the curve along which storage grows and composition gets decided — not about the labor curve of filling a substrate in the first place. What the architecture does claim is that the labor curve is allocable. LLMs can draft substrate content under human governance, and in the POC’s regulated-reporting setting routinely do. Humans can author substrate content directly when stakes or domain knowledge warrant it. Stable cells can be delegated to LLM origination with human inspection and override rights preserved rather than actively exercised. All three modes are available in the same deployment, and humans choose where the labor falls based on stakes and judgment. The infrastructure-cost claim and the labor-cost question are about different things, and keeping them distinct is what lets Claim 4 say what it says and not what it doesn’t.

7. Claim 5 — Tool-agnostic pattern at cell level

7.1 Three minimal requirements; tool-agnosticism at substrate level

Claim 5 states the substrate-layer accessibility commitment that closes Part 1’s core-theory defense. CKS is a tool-agnostic design pattern: any environment meeting three minimal requirements — persistent structured state, human read/write access, and LLM access to substrate content — can instantiate the pattern. This is distinctive because comparable governance-first architectures require specialized runtimes (AI gateways, governance platforms, agentic hubs); “protocol-agnostic” or “platform-agnostic” claims by such platforms refer to agnosticism within their runtime (multiple model providers, agent protocols, or ML backends), not tool-agnosticism at the substrate level.

CKS’s contribution is at the cell level — designing a coordination-knowledge unit non-specialists can govern and maintain in familiar tools, with non-specialist authorship demonstrated by the POC. 2024–2026 research has established commodity spreadsheets as legitimate LLM substrates for execution (TradeSweep, FLAME, SpreadsheetLLM, SheetAgent; Excel Copilot’s =COPILOT() function; third-party =PROMPT() plugins like Cellm), but no work elevates the spreadsheet cell or database record to a first-class governed coordination unit encoding roles, provenance, approval states, and decision rationale. This absence-of-evidence finding is scoped to the four-quadrant landscape §7.2 systematically surveys — spreadsheet-as-execution-substrate research, platform-based AI governance, citizen-developer democratization, and shadow AI empirical reality; we cannot rule out isolated examples not surfaced by this review. CKS extends the recognized “spreadsheets as execution substrate” direction into a governance-first “cell as governed coordination unit” pattern. Cell-to-system composition and coexistence with top-down AI platforms are framed as open questions and long-term vision, not contributions of this paper. Reign — the protocol-agnostic agentic governance platform featured in the iTmethods 2026 vendor comparison (vendor positioning) — is the sharpest concrete instance of the layer distinction the claim’s contrast heads off. Reign’s protocol-agnosticism names its ability to sit in front of multiple agent and tool protocols (MCP, OpenAPI, custom tool APIs) through its AI Gateway and Agentic Hub; that is agnosticism inside a specialized runtime. CKS’s agnosticism operates at a different layer: the substrate can be any environment meeting the three minimal requirements, including environments without any specialized runtime at all. The contrast is not that CKS is more agnostic than Reign; it is that the two architectures are agnostic at different layers — Reign inside its runtime, CKS at the substrate underneath.

7.2 Four-quadrant landscape diagram

The four-quadrant landscape diagram this section anchors arranges the 2024–2026 landscape around Claim 5 into four roughly disjoint categories, none of which occupies CKS’s specific position. Quadrant A — spreadsheet-as-execution-substrate research — holds the peer-reviewed work establishing commodity spreadsheets as legitimate LLM substrates (TradeSweep at ICSE 2025; FLAME from Microsoft Research 2024; TableTalk, SpreadsheetLLM, SheetAgent, and SODBench across 2024–2025 research venues), alongside the native-vendor products that bring in-cell LLM invocation to production (Microsoft’s =COPILOT() worksheet function and “Ask Copilot for a formula,” across Dec 2025 and March 2026 updates) and the third-party plugins that extend the pattern beyond any one vendor stack (Cellm, Numerous.ai, SheetAI, GPT-for-Work). Quadrant B — platform-based AI governance — holds the specialized-runtime vendors (Credo AI, IBM watsonx.governance, Holistic AI, Reign, ModelOp, Lumenova, Securiti; vendor positioning in each case) and the analyst-side taxonomies that carve the vendor space into architectural categories (Trustible’s 16-type taxonomy across five layers, analyst; Ethyca’s eight-platform comparison framed around enforcement depth, analyst/aggregator). Quadrant C — citizen-developer democratization — holds the analyst projections and practitioner framings that treat low-code/no-code platforms as the access route for non-specialists (Forbes essays on the “AI-powered citizen revolution,” practitioner; Gartner projections on citizen-developer growth, analyst; Forrester automation predictions, analyst). Quadrant D — shadow AI empirical reality — holds the bottom-up adoption evidence from a mix of industry research and practitioner commentary (Salesforce 2024 employee survey, industry research; Reco 2025 State of Shadow AI report, vendor data; Forbes Tech Council shadow-AI commentary, practitioner; ABBYY 2026 trends report, vendor trends).

The key observation this section draws from the diagram is structural. CKS occupies the empty

region between Quadrants A and B — a cell-level governance pattern that takes Quadrant A’s commodity-substrate legitimacy as a starting point and extends it to carry the governance semantics that Quadrant B currently locates in specialized runtimes. Quadrant A’s contribution is that commodity spreadsheets and their cells are already recognized as legitimate LLM substrates; the cell in this consensus is an execution unit — a place where formulas run, transformations land, or LLM invocations produce values. Quadrant B’s contribution is that governance content — policies, role assignments, approval states, regulatory mappings, evidence trails — already exists as a coherent architectural concern; the question Quadrant B does not ask is whether that content could live on Quadrant A’s substrates rather than inside specialized runtimes around them. CKS’s architectural move is to ask and answer that question: governance semantics are placed inside the cell, on Quadrant A’s commodity substrate, without requiring any of Quadrant B’s specialized runtime. Quadrants C and D provide the empirical grounding for why the gap matters. Quadrant C documents the demand for non-specialist access to AI-mediated work; Quadrant D documents what happens when that demand is not met inside governed infrastructure — bottom-up adoption inside commodity tools, with top-down responses operating exclusively at the platform level rather than at the cell level where the actual work lives.

The evidence base for the four quadrants is mixed in source type, and the paper labels source types explicitly on first citation rather than flattening them. Quadrant A is mostly peer-reviewed research plus native-vendor products and third-party plugins, each labeled on first mention above. Quadrant B mixes vendor positioning (the runtime vendors themselves) with analyst and aggregator voices (Trustible, Ethyca, iTmethods), each labeled on first mention. Quadrants C and D are predominantly industry reports, vendor data, and practitioner essays: the 55% figure (Salesforce 2024 industry research), the 27% and 400+-day figures (Reco 2025 vendor data), and the ~\$670K incremental breach cost figure (Forbes Tech Council practitioner commentary) are load-bearing for the problem framing but come from sources with known methodological variation. The paper presents those three figures as a cluster demonstrating breadth of evidence across independent reports rather than as a stacked accumulation; the structural claim the cluster supports is that bottom-up adoption inside commodity tools is large, persistent, and consequential, and three independent sources converge on that claim with different measurement methods. Later references to any of these sources do not re-label the source type; first citation carries the label, and the reader can calibrate accordingly.

7.3 Cell as governed coordination unit (not cell as computation); =COPILOT() proximity answered

The distinction the section turns on next is between what the cell is to Quadrant A’s research consensus and what the cell is to CKS. 2024–2026 research has established commodity spreadsheets as legitimate LLM substrates for execution: TradeSweep frames non-specialist natural-language data cleaning as end-user programming over a spreadsheet; FLAME legitimizes spreadsheets as one of the most pervasive programming environments by training a dedicated model on the formulas that inhabit them; SpreadsheetLLM develops LLM interpretation of spreadsheet structure; SheetAgent commands multi-step operations through natural language over sheets. Across this consensus the cell is consistently framed as an execution unit — a locus of formulas, transformations, documentation, or LLM invocations. CKS does not reject that framing and does not claim priority over the commodity-substrate legitimacy those works establish. The paper’s move is narrower and different in kind. CKS makes the cell a governed coordination unit — a cell whose content includes the roles, approval states, authority provenance, and decision rationale that would otherwise live in a special-

ized runtime or be absent altogether. The earlier framing is extended rather than displaced: the substrate the research consensus has legitimized becomes the substrate where governance semantics live as authored content.

The nearest prior-art reference point at the commodity-platform scale is Microsoft’s native =COPILOT() worksheet function, together with the “Ask Copilot for a formula” flow and the third-party =PROMPT()-style plugins that expose LLM invocation through in-cell formulas. The paper answers the proximity question in one sentence that preempts the most likely reviewer misreading: **A governed =COPILOT() call remains a cell-as-computation pattern; CKS makes the cell a governed coordination unit by putting roles, approval states, and decision rationale inside the cell as authored content.** What the sentence does is separate two design objects a careless reading collapses — a cell in which an LLM invocation runs, and a cell whose content encodes governance semantics. The two objects are not on a spectrum of how much governance has been added to a =COPILOT() call; they are different kinds of things. A second implicit-combination candidate is worth naming for the same reason. One might ask whether the combination of in-cell LLM invocation (Quadrant A) and platform-level governance around the runtime (Quadrant B) produces cell-level governance as an emergent consequence — an enterprise-approved =COPILOT() call behaving like a governed coordination unit because it has passed through a governance platform’s approval flow. It does not. The combination gives an enterprise-approved =COPILOT() call; the cell itself is still computation, and the governance is still around the runtime. Nothing in the combination elevates the cell to encode roles, approval states, or decision rationale as authored substrate content. The architectural move CKS makes is not implicitly produced by any combination of existing work in the 2024–2026 evidence base.

7.4 Non-specialist governance (architectural commitment) vs. non-specialist authorship (POC-demonstrated)

The claim’s phrasing about non-specialist access deserves one explicit move on the authority-vs-labor distinction §2.1 introduced and §3.3 rendered portable as *governance is an authority architecture, not a review workflow*. The architectural commitment Claim 5 makes is non-specialist governance — the commitment that a non-specialist can inspect, modify, and override the substrate and the orchestration rules that govern cell execution, in tools they already use. Non-specialist authorship and non-specialist maintenance are different commitments. Authorship is the labor of producing substrate content; maintenance is the labor of keeping it current as the underlying work evolves. CKS’s architecture supports both, and the POC demonstrates both in a regulated-industry setting where the substrate was authored and maintained by non-specialists in commodity infrastructure with a commercially available LLM integration; neither is an architectural requirement the paper claims follows from the design pattern alone. The distinction matters for what the claim is defensible against. A reviewer objecting that the literature does not support non-specialist authorship as an architectural property is correct — that is not what Claim 5 claims. A reviewer objecting that the literature does not support non-specialist governance as a realistic design target is engaged by the literature plus the POC: the literature establishes that commodity substrates are legitimate for non-specialist work, and the POC demonstrates that a non-specialist can in fact author and maintain substrate content in such a setting. The accessibility claim stands on both sources together, not on the literature alone.

One further positioning move situates CKS relative to the closest conceptual umbrella in the 2024–2026 literature. Gargioni et al. (2024), surveying nine papers at the intersection of End-User Development and LLMs (filtered from 103 candidates), propose design implications for future EUD+LLM

environments — conversational stepwise refinement, user control, prompt-authoring tutorials, verification tooling, and meta-design approaches that let non-specialists shape the systems they use. CKS is consistent with all five implications and is offered here as an instance of the kind of concrete substrate architecture those implications call for but do not themselves provide. Gargioni et al. is a survey plus guidelines; it proposes no data model, no substrate definition, no coordination schema, no cell-level semantics. CKS is the concrete design at those layers. Positioning CKS as an instance under the umbrella rather than a competitor to it is the accurate framing — the survey identifies design implications, and the paper contributes a specific architecture that honors them. The pragmatic backing for why the umbrella matters comes from the practitioner literature Quadrant D collected. The “channel, not chase” framing common to recent shadow-AI commentary (practitioner) argues that bottom-up AI adoption in commodity tools is already happening and that the enterprise response should be to channel it into governed infrastructure rather than to chase it into approved-platforms-only compliance. CKS’s architectural commitment to non-specialist governance in commodity tools is the substrate-layer response that framing asks for.

8. Related Work for the Core Theory

8.1 Governance-first architectures

The governance-first architecture cluster visible in the 2024–2026 landscape is internally heterogeneous. MACI (Chang 2026), Stevens and Sakura Sky (2025), and the control-plane architectures each commit to governance as an architectural concern at a specific layer: MACI at the coordination-mechanisms layer (baiting, filtering, persistence); Stevens and Sakura Sky, together with vendor AI gateways such as TrueFoundry’s AI Gateway (vendor), at the control plane (approval gates, reviewer workflows, gateway-routing primitives); and Corvair’s “Architecture-First Governance” writing (analyst/vendor source) at the runtime middleware layer where governance is interposed between agents and enterprise systems. Cognitive Core (Seck 2026) sits differently in this cluster. Seck’s paper is a design methodology for deploying AI systems in regulated contexts rather than an architectural commitment about where governance lives; its three-layer configuration model — workflow YAML, domain YAML, case JSON — is deliberately substrate-agnostic and applies across traditional AI methods, fine-tuned models, and LLM-assisted workflows. CKS joins the cluster with yet a different kind of claim: an architectural commitment about what the substrate is as a design object — its properties, its governance semantics, its position relative to LLM reasoning. These are three different kinds of claim inside the same governance-first cluster, and the cluster is only legible once they are separated.

Positioning CKS relative to this cluster clarifies what kind of contribution the paper is making: a complementary one, not a competitive one. Cognitive Core primitives can execute over CKS cells, with the cell’s governed content supplying the domain knowledge a Cognitive Core deliberation or verification primitive reasons through. Stevens’s control-plane primitives can gate cell-level operations — an approval gate placed between a participant’s proposed cell modification and the cell’s committed substrate state is exactly the kind of composition §4.4 developed. TrueFoundry-style AI gateways can route model calls whose outputs ultimately materialize as substrate content, with gateway-level policy enforcement composing with cell-level governance rather than duplicating it. The governance-first architecture space is not a zero-sum competition among rival designs; it is a multi-layer stack the field is collectively building out, with each contribution claiming its own layer and its own design object. CKS’s contribution is the coordination-knowledge layer and the

substrate-as-design-object move that comes with it.

8.2 Structured-memory and epistemic-substrate proposals

A different 2024–2026 lineage — running alongside the governance-first architectures rather than within them — proposes structured memory or epistemic substrate as the coordination object for AI-mediated work. OIDA (2026 preprint) structures organizational knowledge as typed Knowledge Objects carrying epistemic class (fact, hypothesis, decision, question) and importance scores, with signed contradiction edges between conflicting objects and QUESTION-as-modeled-ignorance as first-class substrate state; a deterministic Knowledge Gravity Engine maintains the substrate. Zahn and Chana’s Knowledge Objects paper (2026) demonstrates the database-like cost model of hash-addressed persistent memory empirically, with $O(1)$ retrieval and constant per-query token cost across corpus sizes from one hundred to one hundred thousand facts. KGLM (Baldazzi et al. 2024) combines a rule-augmented enterprise knowledge graph with chase-based provenance explanations verbalized by an LLM — the same deterministic-substrate-plus-LLM hybrid split §4.1 draws at the governance boundary. Naja and colleagues’ 2021 PROV-O and EP-Plan framework establishes accountability plans and accountability traces as structured governance artifacts, vocabulary §3.1 imported and the paper uses throughout. These works together form the epistemic-substrate lineage CKS extends.

Positioning CKS relative to this lineage is straightforward: each of these works is adjacent prior art on structured, contradiction-aware, or deterministically maintained substrates, not a scoop. Three CKS commitments distinguish the paper across the family. The first is human governance of the substrate as a design requirement rather than an optional attribute of a particular deployment — a commitment about where authority over substrate content sits. The second is formal role and authority schema at the cell level, encoding who may decide what, on what evidence, under which policy. The third is the AI-as-substrate-mediator framing of the LLM’s role: the LLM mediates over human-governed substrate rather than operating as a substrate client reasoning over a structured store. The specific differentiations live in the claim sections themselves: §5.2 develops these three axes against OIDA in detail; §6.2 defends the inheritance-plus-extension framing against Zahn and Chana’s Knowledge Objects. This section’s job is to name the distinguishing commitments as a single family-level statement and point readers toward those treatments, not to repeat them.

8.3 Execution-substrate research

A third 2024–2026 research direction treats commodity spreadsheets as legitimate LLM execution substrates. The research side of this consensus includes TradeSweep, FLAME, TableTalk, SpreadsheetLLM, SheetAgent, and SODBench, each establishing spreadsheets as a first-class target for LLM operations — natural-language-to-formula, table comprehension, agentic manipulation, benchmarking. The vendor side includes Excel’s native =COPILOT() function and “Ask Copilot for a formula,” and the third-party =PROMPT()-style plugin ecosystem — Cellm, Numerous.ai, SheetAI, GPT-for-Work (all vendor sources). Across this quadrant the cell is consistently framed as an execution unit: a locus of formulas, transformations, or LLM invocations that compute values.

CKS’s move in this landscape is to extend rather than reject the research consensus. The recognition that commodity spreadsheets can host LLM-integrated work is the foundation; the paper’s architectural commitment is to extend that recognition into a governance-first pattern where the cell carries not only computed values but roles, approval states, and decision rationale. §7.3 makes the extension move explicit and answers the =COPILOT() proximity question directly; this sec-

tion’s contribution is simply to position the move in Related Work terms — as continuous with the spreadsheet-as-execution-substrate literature rather than in competition with it.

8.4 Where CKS sits — the empty region the paper occupies

The three preceding subsections map three families of 2024–2026 work: governance-first architectures (§8.1), structured-memory and epistemic-substrate proposals (§8.2), and execution-substrate research (§8.3). CKS sits in the intersection of all three. No 2024–2026 work in the paper’s evidence base occupies that intersection: no work argues for human-governed, multi-field-schema, cross-session-addressable, commodity-tool-hostable coordination substrates with AI as substrate mediator. Each family contains work that occupies two of the three, or that overlaps one while committing to a different design object, but the combination — substrate as human-governed design object, multi-field schema carrying rationale, authority, and decision state, persistence and addressability across sessions, hostability on commodity infrastructure, and the LLM as substrate mediator — is the empty region the paper occupies.

The paper’s core-theory contribution is part of an emerging research direction. The CHI 2026 Human-Agent Collaboration Workshop articulates that direction explicitly, calling for the extension of foundational CSCW concepts — articulation work, common ground, workspace awareness — to AI-mediated coordination settings. CKS’s distinctive move within this direction is to locate coordination state in a human-governed, persistent substrate that multiple humans operating in different roles read, write, and audit, rather than in per-session interactions between individual humans and LLM collaborators. The workshop is cited here as alignment with a research direction CKS contributes to, not as a framework the paper instantiates; the paper’s own architectural vocabulary — the LLM as substrate mediator, not as teammate or colleague — marks the specific place CKS sits within the broader direction.

Part 2 — Human-AI-Human Coordination Extension

9. Claim 6 — The human-AI-human coordination extension

9.1 Claim statement — extension of Claim 2’s AI-as-substrate-mediator commitment to multi-human contexts

Claim 6 states the paper’s architectural commitment for the first extension of the core theory — the human-AI-human coordination extension. In the human-AI-human coordination extension of CKS’s core theory, the AI-as-substrate-mediator commitment of Claim 2 extends naturally to multi-human contexts: the cell — aggregating substrates and LLMs operating under the core-theory mediator commitment — mediates coordination across multiple human participants in different roles. Coordination state — the substrate and its cell-level contents — lives outside any individual participant’s session and is readable, writable, and auditable by multiple humans. The AI’s role is to help humans read, write, interpret, and compose substrate content consistently across sessions and participants, while the substrate itself is the source of truth for “what was decided, by whom, under what authority, with what rationale.” The distinctive property is auditable preservation of reasoning and context behind multi-human interactions — not multi-human usage of AI, which existing collaborative tools (Google Workspace with integrated LLMs, Notion AI, Microsoft stack equivalents) already support. This extension draws on recent CSCW-AI work extending articula-

tion work and boundary objects to AI settings (Strauss; Bannon; Star & Griesemer; computational boundary objects in KNIT; articulation work in AI fairness), and on epistemically structured substrate proposals such as OIDA (typed Knowledge Objects with signed contradiction edges and modeled ignorance). It differentiates from these on three axes: (a) explicit commitment to human governance of the substrate as a design requirement, (b) formal role/authority schema at the cell level (who may decide what, on what evidence, under which policy), and (c) AI as mediator across multiple human participants and sessions, rather than AI as consumer of structured memory or assistant to individual users. Based on a systematic review across CSCW research, collaborative AI tools, provenance frameworks, multi-agent coordination literature, and group-chat AI patterns, we find no 2024–2026 work defining AI as mediator of a persistent, human-governed, multi-role coordination substrate; we cannot rule out isolated examples not surfaced by this review. The distinction that matters is between AI-as-assistant (participant-pair interactions, per-session context, AI outputs delivered to one user at a time) and AI-as-mediator (AI operating over shared substrate, cross-session coordination, AI outputs materialized as substrate content all participants can see, challenge, and build on). §4.2 established this mediator commitment as a core-theory commitment that applies to single-human coordination across sessions and flagged its extension to Claim 6; this section extends the commitment into multi-human coordination rather than introducing it here for the first time.

The distinctive property deserves unpacking against the specific landscape of collaborative tooling a reader may have in mind, because it is the line the Related Work for the extension will later defend. What CKS claims is auditable preservation of reasoning and context behind multi-human interactions as a property of a persistent addressable substrate — coordination state that exists outside any individual participant’s session and is readable, writable, and auditable by multiple humans through the cell-level semantics Claim 1 and Claim 3 defended. What CKS does not claim, and what the extension is not about, is multi-human usage of AI — shared access to an LLM from inside a collaborative product. Existing tools already realize that pattern: shared-document platforms with integrated LLMs, workspace tools such as Notion AI, collaborative databases such as Airtable with AI fields, and group-chat AI integrations such as Slack AI and Teams-embedded LLM assistants all embed AI as a per-user assistant or per-document co-creator inside a multi-human product. The data layer of each such product functions as an implicit coordination substrate — decisions, rationale, and contested points do land somewhere in documents, records, or channels — but the substrate is vendor-specific, weak on provenance and authority semantics, and not positioned as a human-governed coordination artifact in the sense Claim 1 defends. The distinction between “coordination state as a persistent addressable object the paper is claiming” and “multi-human access to AI, which existing tools already provide” is the line §9 holds throughout.

The claim statement articulates three differentiation axes; §9.3 develops each in full, but naming them briefly here gives the rest of §9 a shape to refer back to. The first axis is human governance of the substrate as a design requirement — a commitment about where authority over substrate content sits, not a feature of any particular deployment. The second is a formal role/authority schema at the cell level — an explicit encoding of who may decide what, on what evidence, under which policy, carried as substrate content the cell’s orchestration rules draw on. The third is AI as mediator across multiple human participants and sessions — the multi-human rendering of the mediator role §4.2 established, in which AI outputs materialize as substrate entries meant to be seen, challenged, and extended by humans in different roles, rather than as per-session assistance to individual users or as consumption of a structured memory. The three axes are independent architectural commitments, not three framings of the same commitment; §9.3 treats each on its own terms.

9.2 Five-area landscape grouped by analytical affinity

Conceptual and interaction-framing neighbors. The 2024–2026 literature relevant to the extension spans five areas that group into three analytical families admitting tidier positioning than a per-area walkthrough would, and the first family covers work framing how humans and AI relate at the conceptual or interaction level. CSCW-AI work is actively reviving foundational coordination concepts for AI-era settings: computational boundary objects in KNIT’s interdisciplinary co-design tool, articulation-work studies of adapting AI fairness checklists to local practice, grounding-theoretic work on multi-party LLM dialogue, Clark’s common-ground framing as invoked in recent LLM agent research, MindDial’s theory-of-mind treatment of common ground as negotiable shared propositions between agent and user, the 2025 “Building Common Ground” survey noting the absence of a standard representation of common ground for LMs, ThinkFlow’s conversion of multi-turn sense-making conversations into reasoning-aware cognitive workflows, and the CHI 2026 Human-Agent Collaboration Workshop’s explicit call to extend CSCW concepts to AI settings. A second line in this family picks up “mediator” in the interaction sense: the Frontiers 2024 paper on AI as mediator in communication across audiences, and the DeepMind political deliberation work in which AI mediates consensus-building on contested issue framings. A third line treats humans as first-class participants in multi-agent coordination, through human-agent collaboration frameworks, hybrid-pod organizational models, human-in-the-loop patterns for multi-agent systems, “agents in the org chart” writing on support-team deployments, accountability-gap analyses of agent-enabled workflows across sales, legal, finance, and operations, “AI Agents in Program Management” (2026 practitioner), and “Towards AI as Colleagues” (2026 practitioner). The affinity across the three lines is that each is a framing of how humans and AI relate at the conceptual or interaction level — how they coordinate a conversation, a decision, a shared task. CKS’s departure is to locate coordination state in a persistent, human-governed substrate rather than in the interaction pattern, and — for the multi-agent-with-humans line specifically — humans remain gatekeepers, reviewers, or remote collaborators rather than co-authors of a shared substrate. The “AI as teammate” and “AI as colleague” framings prominent in the program-management and colleagues writing are framings CKS departs from, not positions it adopts. The architecture depends on AI being on a different footing from humans: humans author, govern, and hold authority; the LLM mediates over the substrate without holding authority. The extension’s vocabulary is AI-as-substrate-mediator throughout.

Production-tool and instrumentation neighbors. Collaborative AI workspaces are the closest thing in today’s market to a deployed multi-human coordination tool with AI inside it. Shared-document platforms with integrated LLMs, Notion-style workspace tools with AI fields over a unified graph, Airtable-style collaborative databases with cell-level AI outputs, and group-chat AI including Slack AI and Teams-embedded LLM assistants all embed AI as a per-user assistant or per-document co-creator over vendor-specific schemas. Coordination state accumulates in these products as a side effect — documents record what was decided, tickets record who decided it, chat logs record the conversation — but the data layer is high-dimensional and product-specific, and does not carry first-class provenance or authority semantics at the granularity of a coordination decision. Running alongside this production-tool line is a parallel thread in provenance and traceability: PROV-AGENT’s activity-and-agent-level provenance model, the IETF Verifiable AI Provenance Framework draft, clinical AI provenance frameworks, agentic-workflow provenance surveys, and enterprise AI audit-trail writing. These works are converging on graph-structured, evidentiary-grade records of AI operations and decisions, but treat provenance as instrumentation attached to pipelines and agent activity rather than as a foreground human-governed coordination substrate

the humans and the AI both work through. The affinity across the two lines is that each realizes partial coordination capability through production tooling — one by embedding AI in multi-user products, the other by wrapping AI pipelines in traceability infrastructure. CKS’s departure is to make the coordination object primary rather than instrumented around: the substrate is the design object, and provenance and multi-human addressability are properties it carries by construction rather than layers added over an existing pipeline.

Structural-substrate neighbors. A third family of work, closer in architectural spirit, gives the substrate typed epistemic structure as the architectural unit. OIDA is the chief contemporary example: typed Knowledge Objects with epistemic class, signed contradiction edges between conflicting objects, QUESTION-as-modeled-ignorance as first-class substrate state, and the deterministic Knowledge Gravity Engine as maintenance mechanism. Zahn and Chana’s Knowledge Objects paper (2026) is precedent for typed, structured knowledge units as the architectural family OIDA builds on. These works give the substrate the epistemic structure that distinguishes a coordination substrate from a corpus, and they are the closest named neighbors to CKS on the “substrate as design object” dimension. What they do not do — and what CKS’s extension to multi-human coordination requires — is frame AI as substrate mediator across humans. OIDA and Knowledge Objects treat AI as substrate client, consuming typed structure to answer queries or perform classifications, not as mediator coordinating multiple human roles through the substrate. The distinctive position CKS occupies is the intersection of all three families: a human-governed, persistent, low-dimensional coordination substrate with multi-role authority semantics, over which AI — already the substrate mediator in single-human coordination per Claim 2 — mediates across participants and sessions. This is the multi-human instantiation of the core theory, and it is visible in none of the three families in isolation.

9.3 Three differentiation axes

Axis (a) — human governance of the substrate as a design requirement. In current collaborative AI workspaces, schemas are vendor-owned and AI fields and derived outputs are materialized by the model; human governance of the coordination layer is not a design requirement but a feature of whichever product is being used. In CKS, the substrate is human-governed — inspectable, modifiable, and overridable by humans at any time — and that commitment is a design requirement at the architectural level, not an optional attribute of a particular deployment. Authorship may be performed by humans directly or by LLMs operating under human direction; what the architecture commits to is authority, not labor. This is the extension-scope rendering of the same authority-vs-labor distinction Claim 1 defended for single-human coordination: the substrate Claim 1 treats as a human-governed artifact at the coordination and decision layer is the same artifact Claim 6 treats as a human-governed artifact across multiple participants in different roles, and the governance commitment is the same commitment at each scope. §3 defends the substrate-as-human-governed-artifact commitment as core theory, and §3.3’s authority-architecture-not-review-workflow framing applies at the extension scope without modification.

Axis (b) — formal role/authority schema at the cell level. Current tools have access control — who may view or edit which document, which record, which channel — but audit of decision rationales across AI-mediated workflows is ad hoc and spread across documents and tickets, with no cell-level encoding of who may decide what, on what evidence, under which policy. In CKS, every substrate cell carries explicit provenance — who authored or modified the content, under what authority or policy, based on which rationale and evidence — as substrate content the cell’s orchestration rules draw on, and AI is bound by those structures when proposing or recording changes. The

schema is formal in the architectural sense: it is part of what makes the substrate a CKS substrate rather than an arbitrary structured store. §5 defends the corresponding commitment for conflicts — contradictions, inconsistencies, and unresolved ambiguities persisted as first-class addressable objects with provenance attached directly to the conflict itself — and the role/authority schema Claim 6 extends to multi-human contexts is the architectural companion of that commitment: provenance attaches to decisions as it attaches to conflicts, and cell-level orchestration rules operate over both.

Axis (c) — substrate as persistent cross-session object with AI as mediator across participants. Current tools localize AI contributions to individual documents or tasks; AI operates as a per-document co-creator or a per-channel participant-bot, and its outputs live inside the interaction in which they were produced. In CKS, the substrate is a persistent cross-session coordination object that multiple humans in different roles read, write, and audit, and AI operates as mediator over the substrate as a consequence of the core-theory commitment §4.2 defended. The multi-human rendering of that commitment is that AI outputs materialize as substrate entries meant to be seen, challenged, and extended by humans in different roles across sessions, rather than as per-session assistance delivered to one user at a time or as derived metadata attached to a single document. §4 defends the AI-as-substrate-mediator commitment at the core theory, and §7 defends the tool-agnosticism that allows this architecture to be instantiated in commodity infrastructure rather than requiring a specialized multi-human coordination runtime.

9.4 The “architecturally different from OIDA on the multi-human axis, not OIDA plus multi-human” move

OIDA warrants a second treatment at the extension scope because the adjacency it carries at the core-theory scope — typed Knowledge Objects with signed contradiction edges, QUESTION-as-modeled-ignorance, the deterministic Knowledge Gravity Engine as maintenance mechanism, and the general commitment to structured, contradiction-aware, deterministically maintained substrate — is the adjacency most likely to be conflated with the extension itself. §5.2 developed the three-axis differentiation against OIDA in full: OIDA does not commit to human governance of the substrate as a design requirement; it encodes no formal role/authority semantics at the cell level; it frames AI as substrate client rather than as mediator across humans. Those three axes carry over to the extension scope unchanged. The extension does not add a multi-human requirement to OIDA’s design posture; it extends CKS’s already-distinct design posture — human-governed substrate, role/authority schema, AI as substrate mediator — from single-human coordination into multi-human coordination.

The framing that prevents the “OIDA plus multi-human” misreading is the sentence §5.2 landed at the core-theory scope, and it bears repeating here at the extension scope because the misreading is most plausible precisely at the scope where OIDA is silent and CKS is most distinctive: CKS is **architecturally different from OIDA on the multi-human axis, not OIDA plus multi-human**. A reader who encounters Claim 6 first, or who reads §9 as if the extension commitments were what distinguishes CKS from OIDA, may default to the “OIDA plus multi-human” reading unless the phrase is available to deflect it. The phrase is protected across both landing sites because the misreading it preempts is the single most load-bearing risk in the paper’s positioning against epistemically-structured-substrate prior art.

9.5 Epistemic qualifier — systematic review with honest scope limitation

The reframed claim states the epistemic qualifier directly, and it is worth pausing on: based on a systematic review across CSCW research, collaborative AI tools, provenance frameworks, multi-agent coordination literature, and group-chat AI patterns, the paper finds no 2024–2026 work defining AI as mediator of a persistent, human-governed, multi-role coordination substrate; the paper cannot rule out isolated examples not surfaced by this review. The framing is honest scope limitation rather than defensive hedge. The systematic review was substantive: five literature areas scanned, the near-neighbors at each claim’s scope checked for cross-claim overlap, and the evidence weighed against the specific configuration the extension claims. The absence-of-evidence result stands as the finding the review supports. What the paper declines to claim is the stronger assertion that no such work exists at all — an assertion beyond what any systematic review of a moving literature can support. This is the same epistemic stance Claim 5 applied to its empty-region finding in the tool landscape, and the consistency is deliberate: the paper treats absence-of-evidence from systematic review as a bounded, scope-limited claim wherever it is invoked, and the reader is owed that qualification both times.

10. Related Work for the Extension

10.1 CSCW-AI work

The CSCW-AI research direction the extension converses with organizes around three foundational concepts and their contemporary extensions into AI-mediated settings. Strauss’s articulation work names the ongoing organizational alignment labor that coordination depends on, and the 2024 articulation-work-in-AI-fairness studies adapt the concept to analyze coordination performed around AI systems — including the empirical study of practitioners contextualizing a standardized AI fairness checklist to local practice. Star and Griesemer’s boundary objects name the shared artifacts that let differently-situated communities cooperate without full semantic alignment, and KNIT’s computational boundary objects (CHI/CSCW 2026) extend this concept to AI-mediated co-design tools generating adaptive artifacts different experts converge around in real time; the 2024 framing of image-generative AI outputs as boundary objects for participatory design makes the same conceptual extension for generative AI specifically. Clark’s common ground names the dynamic information state jointly managed by participants, and contemporary extensions include MindDial’s theory-of-mind treatment of common ground as negotiable shared propositions, the 2025 “Building Common Ground in Dialogue” survey noting the absence of a standard common-ground representation for language models, and “Learning Partner-Aware Collaborators” (NeurIPS 2025) operationalizing group common ground as measurable propositions grown through reflective prompts. ThinkFlow (CHI/CSCW 2026) sits alongside these as a complementary move, converting multi-turn human-AI sense-making conversations into manipulable reasoning-aware cognitive workflows. The CHI 2026 Human-Agent Collaboration Workshop gathers these threads at the community-direction level. Each of these works supplies either foundational CSCW vocabulary or a contemporary extension of that vocabulary the paper adopts by reference.

The paper positions CKS as a substrate-level architecture the CSCW-AI movement has not itself provided. The foundational concepts — articulation work, boundary objects, common ground — are commitments about how coordination is performed, not architectural commitments about where coordination state lives; the contemporary extensions apply those concepts to AI-mediated settings

without committing to a persistent, human-governed substrate as the object AI mediates. Two comparisons make the move concrete. KNIT scopes computational boundary objects to real-time co-design session convergence; CKS extends the boundary-object concept to cross-session auditable coordination, with the boundary artifact persisting beyond any single session and carrying role and authority semantics the cell’s orchestration rules draw on. ThinkFlow extracts cognitive workflows from sense-making conversations for reuse; CKS positions reasoning-aware artifacts as enterprise-wide coordination substrates, with authored cell-level schema carrying role and authority semantics rather than conversation-derived workflow shape. The CHI 2026 Human-Agent Collaboration Workshop — cited in §8.4 as community-direction alignment for the core theory — extends to the multi-human extension naturally, because the workshop’s call to extend foundational CSCW concepts to AI-mediated coordination settings is precisely the research direction the extension occupies. The detailed differentiations live in §9.3, where the three-axis treatment against the CSCW-AI and collaborative-AI landscape is developed in full; this section’s job is to name CKS’s position within the CSCW-AI movement at the family level rather than to repeat that treatment.

10.2 Collaborative AI tools and group-chat AI patterns

The collaborative-AI-tools landscape is dense and commercially mature, and four product categories frame the contrast directly: shared-document platforms with integrated LLMs, workspace tools with integrated AI over a unified graph (Notion-style), collaborative databases with AI fields (Airtable-style), and group-chat AI including chat-integrated LLM assistants and meeting-AI variants such as Slack AI and Teams-embedded LLM integrations (all vendor products). These tools realize multi-human usage of AI — shared access to LLMs inside collaborative products — which is widely available and increasingly commodified. What the category does not realize, and what the extension names as the missing configuration, is AI-as-mediator of a persistent, human-governed, multi-role coordination substrate: AI’s contributions in current tools are scoped to individual documents, records, or channels, and the coordination state multiple humans would need to read, write, and audit across sessions and across roles lives implicitly in the vendor’s product schema rather than explicitly in a human-governed substrate with role and authority semantics at the cell level.

The architectural difference between current collaborative AI tools and CKS is not a feature gap but a qualitative architectural difference. Current tools are AI-as-assistant over vendor-owned workspaces — AI is embedded in each product’s workspace, operates as a per-user or per-document co-creator, and coordination state, where it exists, lives in the product’s schema. CKS is AI-as-mediator over a human-governed substrate — the substrate is the primary object AI operates over, coordination state persists across sessions and across roles with explicit provenance at the cell level, and AI is bound by the role/authority schema the cell’s orchestration rules enforce. §9.3 develops this difference on the three axes that constitute it, with reference to the collaborative-AI-tool product categories at each axis; this section’s contribution is to name the distinction at Related Work level and point readers toward that treatment, not to repeat it.

Part 3 — Application: Proof-of-Concept

11. Proof-of-Concept Overview

11.1 Regulated industry, commodity spreadsheet, LLM integration

The POC is a cell deployed in a regulated industry, with healthcare as one natural anchor, where coordination runs across multiple functional roles: business stakeholders who carry the domain knowledge, developers who implement the reporting logic, and reviewers who validate output against the regulatory expectations governing the domain. The substrate is hosted in commodity spreadsheet infrastructure with a commercially available LLM integration; no specialized runtime, proprietary governance platform, or bespoke tooling is involved. The substrate is authored and maintained by a non-specialist domain expert — a participant who works inside the coordination workflow rather than one positioned outside it as a tool builder — and it is read from, written to, and audited by participants across each of the roles and across sessions. Industry subsegment, regulatory regime, team structure, and vendor identities are not load-bearing for the architectural claims the paper defends and are not specified; the POC is a real deployment described at the layer where the claims stand or fall.

11.2 Workflow description at the specificity ceiling

The workflow the POC exercises is a regulated-industry reporting workflow in which the three roles coordinate across sessions to produce a structured output — a specification or report that has to satisfy both the business question behind it and the regulatory expectations governing the domain. The coordination happens over five kinds of content the roles produce in different combinations: business descriptions of what a column or metric means; sample values illustrating the intended content; technical column specifications giving datatype, constraints, and lineage; SQL logic translating business definition into executable query; and decision rationale capturing why a particular definition, specification, or query pattern was chosen over the available alternatives. The five content types have to remain recoverable as coordination state — not as recollection, not as scattered artifacts in tickets and documents, but as structured content the substrate carries forward across sessions and across role boundaries.

Without a persistent substrate, the reasoning behind each decision in the workflow is reconstructible only by interviewing participants who were present or by re-reading the ticket and chat history in which the decision was discussed. The outcome of a decision — the business definition that was chosen, the SQL that was written, the column specification that was approved — typically survives in the document or database where the final artifact lives; the reasoning does not. When a later participant extends the work, a reviewer audits it, or the same author returns to it in a new session, the rationale has to be recovered indirectly, often imperfectly, and sometimes not at all. The POC’s design hypothesis is that a human-governed substrate organized around typed cells can carry that reasoning as addressable content rather than leaving it embedded in the conversational trail that produced it. §11.3 describes what the POC built to test the hypothesis.

11.3 What was built

The substrate is a dual-view structured artifact whose cells carry the five content types together — business descriptions, sample values, technical column specifications, SQL logic, and decision rationale — as addressable, typed content. The structure is hybrid by commitment: the substrate

handles coordination and governance, and the LLM integration handles the reasoning operations that happen over the substrate’s content; the division between them is the governance boundary the core theory draws. Conflicts between business definitions, between a specification and its rationale, or between competing SQL patterns are preserved in the substrate as first-class addressable content rather than auto-reconciled — the substrate does not silently merge or discard them, and the LLM integration cannot resolve them outside what human-authored orchestration rules allow. Authority over the substrate’s content, structure, and orchestration rules stays with the humans who work inside the workflow; content may be drafted by the LLM integration under human direction or authored directly, but the architecture preserves the authority to inspect, modify, and override. The LLM integration is the substrate mediator — it assists participants with lookup over prior rationale, drafting of new content, and interpretation of existing cells — and the infrastructure is commodity spreadsheet tooling that supports persistent structured state, human read/write access, and LLM access to substrate content, which is what tool-agnosticism at the substrate layer requires. Adding cells or expanding content grows storage linearly; no parameter growth, re-embedding, or agent-coordination overhead is involved.

The human-AI interaction pattern the POC instantiates is the one §9 defended as the multi-human extension of the core theory. The non-specialist domain author writes substrate content — business definitions, specifications, rationale — directly, and the LLM integration assists with lookup across prior cells, drafting of new content under the author’s direction, and interpretation of existing content when a question about meaning arises. Participants in each of the roles read, write, and audit the same substrate: the business stakeholder reads a technical specification and writes rationale against it; the developer reads the rationale and writes SQL logic that respects it; the reviewer reads all three and writes audit notes that the next session inherits. No role has a privileged view of the substrate that the other roles cannot see; no role’s contributions are hidden behind a tool-level wall from the other roles’ inspection. This is the concrete realization of what §7.4 framed as non-specialist governance in commodity infrastructure with non-specialist authorship demonstrated alongside it, and what §9 framed as the human-AI-human coordination extension — the substrate the core theory describes, instantiated with multiple human participants in different roles, yielding the additional capability the extension names.

12. How the POC Demonstrates CKS

12.1 Integrated demonstration across all six claims — claim-level narrative

The POC is one deployment, not six. The paper’s six claims are six architectural propositions about the CKS design pattern, but the POC is not six separate instances validating one claim each; it is one cell whose operation exercises the core theory and the human-AI-human extension simultaneously, because the same substrate that supports a single participant coordinating across sessions also supports multiple participants coordinating across roles. The integrated character of the demonstration is structural, not coincidental: what makes the substrate work for a solo author across sessions — persistent coordination state, human authority over content, LLM assistance mediated by the substrate rather than routed around it — is exactly what makes it work across multiple roles, because no role is architecturally privileged over the others. The POC does not demonstrate six things in six ways; it demonstrates one pattern in one way that happens to exhibit all six claims’ architectural properties under operation.

The substrate as it runs in the POC is a human-governed coordination artifact at the coordination and decision layer, which is what Claim 1 commits the paper to — the humans who work inside the workflow hold authority over the substrate’s content, structure, and orchestration rules, and they exercise that authority both directly (by editing substrate content) and indirectly (by writing the rules that govern how the cell behaves when the LLM integration operates over it). Rule conflicts that arise inside the workflow — a business definition inconsistent with an earlier decision, a specification that contradicts an existing SQL pattern — surface in the substrate as addressable content rather than dissolving into the conversational log, which exercises Claim 3’s commitment to conflict as a first-class object, though on the rule-conflict instances the workflow naturally produces rather than as a scaled stress test. The POC’s cell schema — the typed-cell structure carrying the five content types in coordinated form — is the material instantiation of Claim 2’s governance boundary and schema-level decomposition, because the schema is what determines where governance content lives and how the coordination knowledge is partitioned. The linear-cost commitment Claim 4 defends is inherited by the POC from the commodity infrastructure the substrate runs on — the cost model is a property of the spreadsheet substrate, not something the POC independently validates at scale. The POC’s existence in commodity spreadsheet infrastructure with a commercially available LLM integration is the existence proof Claim 5 needs — not a new tool, not a new platform, but the pattern instantiated where non-specialists already work. The multi-role cross-session use of the substrate — business stakeholders, developers, and reviewers each reading, writing, and auditing the same artifact across sessions — is what Claim 6’s human-AI-human extension describes in operation. The narrative here is framing; the audit is §12.2.

12.2 Honest-limits audit — which commitments are strongly exercised, which are partially exercised, which are inherited not demonstrated

The POC’s relationship to each of the six claims varies by exercise strength, and the paper is explicit about which commitments the POC demonstrates robustly, which it exercises only partially, and which it inherits from the substrate infrastructure without independently validating.

Strongly exercised: - Substrate as human-governed coordination artifact (Claim 1) - Schema-level decomposition at the cell level (Claim 2) - AI as substrate mediator in single-human use (Claim 2) - Tool-agnosticism in commodity spreadsheet infrastructure (Claim 5) - Non-specialist authorship as POC-demonstrated capability (Claim 5) - Multi-role cross-session coordination (Claim 6)

Partially exercised: - Conflict as first-class addressable object (Claim 3) — exercised on the rule-conflict instances the workflow naturally produces, not as a scaled stress test - Two-level conflict handling (Claim 3) — cell-level orchestration rules are exercised, but the depth of orchestration-rule variation is limited by the scope of the workflow the POC runs

Inherited not demonstrated: - Linear-cost model (Claim 4) — the cost model is inherited from the substrate infrastructure (Knowledge Objects empirical validation at $\sim 252\times$), not independently validated by the POC at large N

These are the places where a future empirical paper would strengthen CKS; this position paper does not claim to strengthen them.

12.3 Two axes of benefit — parallel findings, not separate claims

The two axes of benefit the POC demonstrates — individual-author coordination across sessions, and multi-human coordination across roles — are parallel findings from the same substrate, not

separate claims about different architectures. One architecture, two modes of use. The individual-author benefit and the multi-human coordination benefit both arise because the substrate is persistent, human-governed, and LLM-mediated in the way the core theory specifies; neither mode requires the substrate to be anything other than what the core theory describes. The POC is one deployment exhibiting both modes, and the analytical contribution of §12.3 is to name them as parallel findings rather than to treat the multi-human mode as a separate contribution stacked on top of the individual-author mode.

The individual-author benefit is what the core theory pays off even for solo users. A single participant working across sessions on a regulated-industry reporting task accumulates reasoning — decisions about business definitions, rationale for specification choices, SQL patterns chosen over alternatives, conflicts that were and were not resolved — and without a substrate that reasoning lives in per-session context, ad hoc documents, or the participant’s memory. With the substrate in place, the same participant returning to the work a week later gets LLM-assisted retrieval and interpretation over the prior rationale, the existing schemas, and the logic that was written before; the substrate is the coordination state that makes the continuity possible. The value holds when no other human ever participates — the core theory describes a complete pattern at the single-human scope, and the POC shows the pattern paying off at that scope before any multi-human capability is invoked.

The multi-human coordination benefit is the additional capability the same architecture supports when multiple roles are present, not a separate feature the architecture acquires for multi-human use. The same substrate that carries prior rationale for a solo author carries it across role boundaries for participants who are not the same person: the business stakeholder, the developer, and the reviewer each read, write, and audit the same human-governed artifact, and no role has a privileged view of it the others cannot see. Because no role-specific view of the substrate has primacy, cross-role coordination is what the substrate yields when the roles are populated; the substrate is not reconfigured for the multi-human case. This is what makes Claim 6’s extension defensible as an extension rather than as a different theory — the same architecture, yielding additional capability when additional participants are present, not a new architecture erected to make multi-human coordination possible.

Closing

13. Discussion and Implications

13.1 Design-pattern contribution vs. empirical contribution

The paper’s contribution is a design pattern, not an empirical evaluation. The paper articulates CKS as a unified design pattern with six architectural commitments, defends those commitments through six claims supported by systematic review across 2024–2026 literature, and instantiates the pattern in one working POC in commodity infrastructure. The POC grounds feasibility; it does not ground generality, and the paper does not make a benchmark claim. One calibration on Claim 4 is worth naming explicitly at this point in the paper rather than leaving it inside §6. CKS inherits rather than independently demonstrates the linear-cost model empirically validated by Knowledge Objects (Zahn & Chana 2026), and extends it on two axes — richer governance semantics, and authored-schema disambiguation rather than algorithmic disambiguation — treated in full

at §6.2 ¶3. Inheritance-plus-two-axis-extension is the accurate positioning; the POC demonstrates a working instance of the extended pattern without independently validating either extension axis at scale, which is what §12 is describing when §12.2 lists the linear-cost model as inherited rather than demonstrated.

What the paper is not follows from what it is. It is not a new system, not a vendor product, not a novel algorithm, not an empirical study. The adjacent work — design-pattern writing in CSCW, software architecture, and information-systems research — is intellectual lineage the paper draws on rather than competition it displaces. CKS is a pattern in the sense those traditions use the term: an articulation of a recurring structure of design problem and design response, named so that subsequent work can recognize it, adopt it, vary it, compose it with adjacent patterns, or argue against it. The paper’s genre is design-pattern writing; its contribution is what it adds to that lineage at the coordination-knowledge layer.

13.2 Scope and limits

The scope limits are real and worth naming in one place rather than leaving distributed across the claim sections. The paper stands on one POC in regulated-industry reporting, and there is no cross-domain empirical breadth; the POC’s weight in the overall argument is bounded accordingly. Cell-to-system composition — how cells aggregate into an organizational-scale coordination fabric — is framed at §13.3 as open question and long-term vision, not as a contribution of this paper. Coexistence with top-down AI platforms is future work; the paper locates CKS within the broader governance-first stack at §8.1 but does not defend a deployment story for CKS cells running alongside runtime governance middleware. On Claim 4 specifically, the inherited-vs-demonstrated gap flagged in §13.1 ¶1 is also a scope limit: CKS commits to the design constraints that produce the linear cost curve and names the curve’s empirical anchor in Knowledge Objects, but the curve itself is not independently measured in a CKS deployment. These are limits of the genre. A design-pattern paper defends architecture and shows instantiation; scaled cross-domain validation is what a successor empirical paper adds.

The evaluation gap deserves a separate calibration because its honest framing is field-level rather than CKS-specific. No benchmark exists for what could be called governance-grade substrate evaluation — evaluation that measures how well a substrate supports cross-session, cross-role coordination under human governance, as distinct from retrieval accuracy, generation quality, or task completion. The systematic-review methodology behind Claim 1 (Elicit report) flagged this as a field-level limit, not a CKS-specific deficit; no work in the evidence base supplies a benchmark the paper could adopt wholesale or that a successor empirical contribution could inherit ready-made. CKS inherits the gap rather than creating it. The POC partially fills it — regulated production use is a form of evaluation, and the claim sections draw on the POC where architectural commitments pay off in practice — but the POC does not constitute a benchmark, and the paper does not claim that it does. What a future empirical contribution would add is the benchmark itself: a measured evaluation protocol against which CKS deployments and other governance-first substrates could be compared directly.

13.3 Open questions for future work

Four open questions frame the paper-level future-work agenda, and none is defended here. The first is cell-to-system composition: how individual cells aggregate into an organizational-scale coordination fabric, how composition is governed when many cells touch overlapping domains, and what

the architectural primitives for that aggregation look like. The second is coexistence with top-down AI platforms — CKS cells deployed alongside runtime governance middleware of the kind Reign, Credo AI, and comparable vendors provide, where each layer governs what it is suited to and the composition between them is made explicit. The third is substrate interoperability across organizations, including substrate-as-regulatory-artifact and substrate-based institutional memory: the substrate as the durable carrier of coordination knowledge across organizational or jurisdictional boundaries. The fourth is substrate-mediated human-AI-agent coordination, in which AI agents operate over human-governed substrates as the coordination artifact rather than over per-session contexts or bespoke agent memory. Each of these is an extension theory the paper acknowledges; each would stand on its own claim set rather than on arguments this paper develops.

Of these directions, evaluation is the most actionable. A successor empirical paper could strengthen three specific claims at the points where the current paper inherits or partially exercises rather than independently demonstrates. Claim 1 would strengthen under a deployment study measuring non-specialist governance at scale — across multiple deployments, multiple domains, and participants who are not the original authors. Claim 3 would strengthen under a stress test of conflict-as-addressable-object at higher conflict density than any single workflow naturally produces, pushing on the limits of orchestration-rule authoring. Claim 4 would strengthen under independent cost-curve measurements taken in a CKS deployment rather than inherited from Knowledge Objects, closing the inheritance-vs-demonstrated gap §13.1 ¶1 flagged. Each direction is complementary to the current paper, not corrective of it: the architecture is defended here; scaled measurement is what the successor adds. One housekeeping note belongs here. Freshness has been named as a desideratum for governed institutional knowledge (Atlan 2026, industry source), but CKS’s four claim-level design goals are traceability, auditability, conflict preservation, and governance; freshness is not among them. The architecture supports freshness in practice — humans and LLMs under human direction can update substrate content at any time — without the paper defending freshness as a claim-level commitment.

Conclusion

CKS is a design pattern in which coordination knowledge, governance semantics, and conflict-handling logic live in a persistent, human-governed substrate outside the LLM. The core theory rests on six architectural commitments defended through five claims; the human-AI-human coordination extension is defended through Claim 6. Both are demonstrated through a single POC in a regulated industry, instantiated in commodity spreadsheet infrastructure with a commercially available LLM integration. The unifying move the paper has defended throughout — outside-the-model vs. inside-the-model — names where coordination, governance, and decision state live in this design pattern, and where they do not.

The paper offers one contribution to an emerging conversation across three research directions: CSCW-AI, governance-first AI architectures, and epistemically-structured substrates. The paper does not claim to close any of these conversations. What CKS adds is a substrate-level design pattern articulated with its architectural commitments, its core-and-extension structure, and a working instantiation in commodity infrastructure — a pattern available to be adopted, varied, composed with adjacent patterns, or argued against. CKS sits at the coordination and decision layer and composes with reasoning-execution, coordination-mechanism, and control-plane layers rather than replacing them. That is where the paper places CKS, and what it claims for it.

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This references list is partial. Additional entries — including vendor documentation, analyst reports, practitioner essays, and preprints requiring further verification — will appear in a revised version.